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Multivariate portfolio choice via quantiles

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We first show how the quantile approach used for univariate optimal portfolio choice can also be useful when dealing with the multivariate case. Specifically, when a related multivariate risk-sharing problem (in the absence of a financial market) can be solved explicitly, then the multivariate optimal portfolio choice is shown to reduce to a one-dimensional problem that can be dealt with using the quantile approach. We then use this finding to develop an efficient algorithm to determine optimal portfolios. We also develop a numerical approach that makes it possible to obtain approximate solutions for general multivariate portfolio selection problems.

Keywords: Decision analysis; Multivariate optimal portfolio choice; Quantile approach; Multivariate utility; Cost-efficiency

JEL Classifications: C02, G11, D81, C61

1. Introduction

A number of papers have successfully used the so-called quantile approach to solve univariate optimal portfolio choice. Carlier and Dana (2006) solved the law-invariant concave utility. Xu (2016) employed the quantile formulation for solving the case of rank dependent expected utility, whereas Jin and Zhou (2008), He and Jiang (2021) and Boudt *et al.* (2022) dealt with portfolio choice under Yaari's dual theory of choice (Yaari 1987). For more references on portfolio choice using the quantile formulation (also under different purposes), we refer to, for instance Carlier and Dana (2011), He and Zhou (2011), Zhang *et al.* (2011), Bernard *et al.* (2014, 2015, 2018), Rüschendorf and Vanduffel (2020) and references therein.

The aforementioned approach builds on the elegant concept of cost-efficiency originally proposed by Dybvig (1988a, 1988b); that is, how to achieve the minimum cost of a payoff (or strategy) for a given distribution of terminal wealth. It readily shows that optimising an increasing law-invariant objective function (e.g. an expected utility) amounts to considering in the optimisation only cost-efficient payoffs, i.e. payoffs that are decreasing with the state price density. As these payoffs are fully characterised by their quantile function, the optimisation problem reduces to a search for an optimal quantile function.

However, all the above-mentioned references deal with portfolio selection problems in a univariate setting. In this paper, we propose to use the quantile approach to solve multidimensional portfolio problems. Specifically, we consider a multivariate law-invariant objective function $V(\cdot)$ that we aim to optimise over a set $\mathcal{A}$ of multivariate payoffs $(X_1, \dots, X_d)$ under a budget constraint. To solve this problem, we relate it in a first step to a multivariate allocation (risk sharing) problem in which the distribution of the total portfolio $\sum_i^d X_i$, $i = 1, \dots, d$, is given and where we aim to find the optimal multivariate allocation that maximises $V(X_1, \dots, X_d)$. In a second step, we then show how the quantile approach can be employed.

Multivariate allocation problems have been extensively studied in the literature. Landsberger and Meilijson (1994) investigated Pareto-optimal risk allocations with respect to convex order and showed that comonotonicity is a necessary and sufficient condition for Pareto optimality. These authors also provide an algorithm (valid in two dimensions) that yields for every non-comonotone allocation a comonotone one that is better (see also Chateauneuf *et al.* 2015 for a different computational solution). Further extensions of the results of Landsberger and Meilijson (1994) are stated in Ludkovski and Rüschendorf (2008), Carlier *et al.* (2012) and Denuit *et al.* (2023). Chateauneuf *et al.* (2000) examine the Pareto-optimal risk sharing and equilibrium for Choquet expected utility functions, showing that optimal allocations are comonotone. Embrechts *et al.* (2018) solve the Pareto-optimal risk sharing problem among agents when

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the objective function is described by the Range Value-at-Risk (RVaR). Other relevant results can be found in Rüschendorf (1995), Cuesta-Albertos *et al.* (1997), Barrieu and El Karoui (2005), Acciaio (2007), Filipović and Svindland (2008), Jouini *et al.* (2008), Tsanakas (2009), Dana and Le Van (2010), Ekeland *et al.* (2012), Mastrogiacomo and Rosazza Gianin (2015) and references therein.

Note that the general problem formulation mentioned above covers different practical decision-making settings. As a matter of fact, several economic interpretations of X_i can be considered. For instance, each X_i can represent the portfolio or performance of trader i . A supervisor or risk manager within a trading firm can be interested in aggregating the performances of individual traders into a global performance criterion. Such aggregation implemented by the supervisor/risk manager can take the form of a multivariate utility function that depends on each X_i . Alternatively, the supervisor/risk manager can consider a multivariate risk measure, such as minimising the joint probability of exceeding the Value-at-Risk thresholds. Another possible scenario is the collective decision-making setting. Consider a fund manager responsible for a portfolio funded by d individuals, each contributing an amount X_i . The fund manager's challenge is to construct a portfolio that maximises a collective performance criterion, taking into account the combined preferences of all investors. Another setting in which multivariate portfolio optimisation can emerge is corporate capital budgeting or multi-division investment allocation. Imagine a corporation with d business units or subsidiaries. Each unit is responsible for a specific investment or project with return X_i . In this case, the decision-maker (e.g. the Chief Financial Officer or an investment committee) must allocate a total budget across these divisions. For instance, the decision-maker could be interested not only on maximising the objective function (e.g. weighted sum of expected returns) but to take into account risk preferences, strategic proprieties or regulatory considerations specific to each unit. We want to highlight that, while these scenarios motivate the use of multivariate objective functions, our focus remains on the methodological framework for solving multidimensional portfolio problems. The specific choice of objective function depends on the application context and is beyond the scope of this work.

Many studies examined portfolio optimisation under quantile preferences; i.e. investors who aim to maximise a fixed quantile of the portfolio returns. For instance, Giovannetti (2013) examines a two-period economy for an asset pricing model based on quantile preferences. de Castro and Galvao (2019) developed a dynamic model of rational behaviour under uncertainty where the agent maximises the stream of future quantile utilities. de Castro and Galvao (2022) provided axiomatisations for both static and the recursive quantile preferences and de Castro *et al.* (2022) study a static portfolio selection problem in which an agent maximises the quantile of portfolio, either between two risky assets or between a risk-free asset and a risky asset. He *et al.* (2025) investigated dynamic portfolio selection in which they considered an agent who continuously trades a risk-free asset and multiple risky assets over time for maximising a quantile (particularly under median maximisation) of terminal wealth at a fixed investment horizon. In our framework, the quantile is used as a

tool (i.e. the quantile formulation) to solve optimal portfolio choice problems under general law-invariant objective functions (e.g. expected utilities, distorted expectations, and can be also quantile functions). Specifically, the search for an optimal payoff, which is a random variable, is reduced to finding its optimal distribution, which is represented as the quantile function.

This paper makes the following contributions. First, we show that when a multivariate risk-sharing problem has an explicit solution (for any given distribution of the total risk), then the corresponding multivariate portfolio selection problem reduces to a one-dimensional portfolio problem in which the quantile approach applies. Second, based on this observation, we develop a numerical approach (Section 3) allowing to solve the multivariate portfolio problem, hereby extending the work of Bernard *et al.* (2019) who dealt with the one-dimensional setting. Third, we also propose a crude method (Section 5) that can solve the multivariate portfolio selection problem when the corresponding multivariate risk sharing cannot be solved explicitly, and when the dimensionality of the problem is small to moderate.

The rest of the paper is organised as follows. The portfolio problem, the multivariate risk sharing problem and the quantile formulation are presented in Section 2. The numerical method building on the quantile approach is discussed in Section 3 and Section 4 examines some numerical evidence within the framework of expected utility theory. Section 5 is dedicated to the general numerical method (that is, when no explicit or semi-explicit solutions are available for the multivariate risk-sharing problem). The section presents some numerical analyses for the expected utility framework, demonstrating the accuracy of our approach and then in a distorted expectation setting, thereby illustrating its potential for use in settings beyond the expected utility environment (Section 5.2). Section 6 provides examples addressed using the method introduced in Section 5 that, to the best of our knowledge, cannot be solved explicitly and for which no other approach seems to be available in the literature. Finally, Section 7 concludes.

2. Multivariate portfolio problem

In this section, we briefly state the assumptions regarding the financial market, present the problem and show how it can be formulated as an optimisation over quantiles.

2.1. Optimal portfolio problem

Let us assume a frictionless and arbitrage-free financial market living on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is a non-empty sample space, $\mathcal{F}$ is the σ -algebra generated by Ω and $\mathbb{P}$ denotes the probability measure on Ω (see Björk 2019 for further details on arbitrage-free financial markets). Let $\mathcal{X}$ be the set of random variables with finite first moments on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We consider a fixed investment horizon $T > 0$ without intermediate consumption in which a final payoff X_T received at time T has an initial price given as

$\mathbb{E}[\xi_T X_T]$ where ξ_T is the pricing kernel, agreed by all agents, with positive density on $\mathbb{R}_+ \setminus \{0\}$.

Our objective is to solve a multivariate optimal portfolio choice problem in which we optimise a multidimensional objective function $V(\cdot)$. We assume that $V(\cdot)$ is law-invariant (that is, if $(X_1, \dots, X_d)$ and $(Y_1, \dots, Y_d)$ are equal in distribution, then $V(X_1, \dots, X_d) = V(Y_1, \dots, Y_d)$). Furthermore, we assume that $V(\cdot)$ is strictly increasing in at least one of the d dimensions. Without loss of generality, we can thus assume that for any constant $a \in \mathbb{R}_+ \setminus \{0\}$, $V(X_1 + a, X_2, \dots, X_d) > V(X_1, X_2, \dots, X_d)$. Finally, we assume that the general portfolio problem (see Problem 2.1 hereafter) is well-posed in that there exists an optimal multivariate portfolio $(X_1^*, \dots, X_d^*)$, leading to a maximum finite value for $V(X_1, \dots, X_d)$.[†]

Note that, for the ease of reading, we usually omit the notion of time. Thus, throughout the rest of the paper, every X_i must be interpreted as $X_{i,T}$, for $i = 1, \dots, d$ (that is, they share the same investment horizon T). The goal of the paper is to optimise a multivariate law-invariant objective function $V(\cdot)$ over a set of admissible $(X_1, \dots, X_d) \in \mathcal{A}$ such that the total budget w_0 is allocated. Formally, we obtain the following problem formulation.

PROBLEM 2.1 *Let $V(\cdot)$ be a multivariate law-invariant objective function. We consider the following multivariate portfolio problem*

$$\sup_{(X_1, X_2, \dots, X_d) \in \mathcal{A}} V(X_1, X_2, \dots, X_d), \quad (1)$$

where $\mathcal{A} = \{(X_1, X_2, \dots, X_d) \in \mathcal{X}^d \text{ s.t. } \mathbb{E}[\xi_T \sum_{i=1}^d X_i] = w_0\}$ and $w_0 > 0$ denotes the total budget that must be allocated.

Since (1) is a one-period optimisation problem with a fixed investment horizon $T > 0$ and does not involve re-balancing or intermediate consumption, the optimal multivariate portfolio $(X_1^*, X_2^*, \dots, X_d^*)$ represents a pre-committed strategy.

2.2. Multivariate risk-sharing problem

To solve the general multivariate portfolio problem (1), we first solve a corresponding multivariate risk-sharing problem in the absence of a financial market.

For each random variable S , let the set $A_d(S)$ be defined as

$$A_d(S) := \left\{ (X_1, X_2, \dots, X_d) \in \mathcal{X}^d : \sum_{i=1}^d X_i = S \right\}. \quad (2)$$

The optimal multivariate risk-sharing problem associated with (1) is written as

$$\sup_{(X_1, X_2, \dots, X_d) \in A_d(S)} V(X_1, \dots, X_d). \quad (3)$$

[†] This is a strong assumption but it seems not possible to give straightforward and verifiable conditions on $V(\cdot)$ and ξ_T to ensure this well-posedness in all the cases covered in the paper. Indeed, we note that even in the univariate portfolio maximisation, the conditions, ensuring the well-posedness for all law-invariant increasing objective function $V(\cdot)$, are not known. Specific conditions for the expected utility maximisation are derived in Jin *et al.* (2008) and for the optimal portfolio of a Cumulative Prospect Theory investor in Jin and Zhou (2008).

When an additive multivariate utility function is used, i.e. when $V(X_1, \dots, X_d)$ is of the form $V(X_1, \dots, X_d) = \mathbb{E}[\sum_{i=1}^d k_i U_i(X_i)]$ in which U_i , $i = 1, \dots, d$, are either exponential utility functions or constant relative risk aversion (CRRA) utility functions, the multivariate risk-sharing problem (3) is solved explicitly in some cases in Gerber and Pafumi (1998, Section 8, p. 80). Specifically, they solve the case in which all risk aversion coefficients are the same and $k_i = 1$, $i = 1, \dots, d$. We provide a modest extension of their results by providing a solution in the case of heterogeneous risk aversion coefficients (see Appendix 2). Further examples are studied in Section 4.

2.3. Quantile formulation

Proposition 2.1 shows how to construct a solution to the multivariate optimal portfolio choice problem (1) provided that a solution to the risk sharing problem (3) can be found.

PROPOSITION 2.1 *Quantile Formulation Let $V(\cdot)$ be a d -dimensional objective function that is law-invariant and strictly increasing in at least one dimension. Let S be any given random variable and assume that the associated risk-sharing problem (3) has an explicit solution $(Y_1(S), \dots, Y_d(S))$. Let $H_T = 1 - F_{\xi_T}(\xi_T)$, then the solution to problem (1) is written as*

$$(Y_1(F_S^{-1}(H_T)), \dots, Y_d(F_S^{-1}(H_T))) \quad (4)$$

in which F_S^{-1} is the quantile function[‡] that maximises

$$\sup_{F_S^{-1} \text{ s.t. } \mathbb{E}[\xi_T F_S^{-1}(H_T)] = w_0} V(Y_1(F_S^{-1}(H_T)), \dots, Y_d(F_S^{-1}(H_T))). \quad (5)$$

Proof Using the main results of Bernard *et al.* (2023, Theorems 3.1 and 3.2), we know that an optimal solution $(X_1^*, X_2^*, \dots, X_d^*)$ to (1) is such that $\sum_{i=1}^d X_i^*$ and ξ_T are anti-monotonic. Let F_S denote the cdf of $\sum_{i=1}^d X_i^*$, then $\sum_{i=1}^d X_i^* = F_S^{-1}(H_T)$ because H_T is decreasing in ξ_T . Furthermore, observe that for any given random variable S , by assumption, we solve the multivariate risk-sharing problem (3) in the absence of the financial market, i.e. the explicit solution $(Y_1(S), \dots, Y_d(S))$. Therefore, the only unknown in the multivariate problem is the distribution of the random variable S and thus the formulation (5) follows. ■

Proposition 2.1 allows us to reduce the dimensionality from d dimensions to one for any multivariate portfolio problem for which the multivariate risk-sharing problem without a financial market is solved explicitly. Thus, finding a solution to (1) boils down to solving two problems: (i) a multivariate risk-sharing problem (3) and (ii) a one-dimensional quantile optimisation problem (5). We use this observation to propose in Section 3 a method for obtaining solutions. This numerical approach, which we call the Multivariate-to-Univariate (also labelled MtU) approach, is fast and accurate, but it requires the availability of a solution of the risk-sharing problem (3).

[‡] Let F_X denote the cumulative distribution function (cdf) of a general random variable X , then the quantile function of X , for $p \in (0, 1)$, is defined as $F_X^{-1}(p) = \inf\{x \in \mathbb{R} : F_X(x) \geq p\}$.

3. Multivariate-to-Univariate (MtU) approach

In situations in which for any given risk S , there is an optimal solution to the corresponding multivariate risk-sharing problem (3), the remaining unknown is the quantile F_S^{-1} that solves (5). To do so, we get inspiration from the numerical approach developed by Bernard *et al.* (2019, Section 3).

We first discretise the problem. Thus, we define $\vec{\xi} := (\xi_i)_{i=1}^n$ in which

$$\xi_i := F_{\xi_T}^{-1} \left(\frac{n+1-i-0.5}{n} \right), \quad \text{for } i = 1, 2, \dots, n,$$

where $F_{\xi_T}^{-1}$ denotes the quantile function of ξ_T which is, by assumption, continuously distributed. Thus, by construction, $\xi_1 > \xi_2 > \dots > \xi_n$. Similarly, if S is ‘approximated’ by the vector $\vec{S} = (s_i)_{i=1}^n$ taking n equiprobable values $s_1, s_2, \dots, s_n$, we can rewrite the optimisation problem (5) in the following way

$$\sup_{(s_1, s_2, \dots, s_n) \in \mathcal{B}} f(s_1, s_2, \dots, s_n), \quad (6a)$$

for the appropriate choice of $f(\cdot)$ that depends on $V(\cdot)$ [†] and where the admissible set $\mathcal{B}$ is now defined as

$$\mathcal{B} := \left\{ (s_1, s_2, \dots, s_n) \in \mathbb{R}^n \mid \frac{1}{n} \sum_{i=1}^n \xi_i s_i = w_0 \text{ (budget constraint)} \right\}. \quad (6b)$$

Because of cost-efficiency, an optimal solution $\vec{S}^*$ must exhibit anti-monotonicity with $\vec{\xi}$, i.e.

$$s_1^* \leq s_2^* \leq \dots \leq s_n^*.$$

Let us introduce the auxiliary vector $\vec{Y} := (y_i)_{i=1}^n$, $y_i = s_i - s_{i-1}$, with the convention that $s_0 = 0$. Given that optimality requires that the s_i , $i = 1, \dots, n$, must be increasing, it holds that

$$y_1 \leq y_1 + y_2 \leq \dots \leq y_1 + y_2 + \dots + y_n = s_n$$

where $y_i \geq 0$ for all $i = 1, \dots, n$. The optimisation problem (6) can be rewritten as

$$\sup_{(y_1, y_2, \dots, y_n) \in \mathcal{C}} f(y_1, y_1 + y_2, \dots, y_1 + y_2 + \dots + y_n), \quad (7a)$$

where the admissible set $\mathcal{C}$ is

$$\mathcal{C} := \left\{ (y_1, y_2, \dots, y_n) \in \mathbb{R}_+^n \mid \sum_{j=1}^n \xi_j y_j = w_0 \right\} \quad (7b)$$

and $\zeta_j := \frac{1}{n} \sum_{k=j}^n \xi_k$ for $j = 1, \dots, n$. As discussed in Bernard *et al.* (2019), the discretisation makes it possible to have an optimisation problem under linear constraints in which the admissible set is reduced to the first orthant by the transformation from (6b) to (7b). In addition, the formulation of the optimisation problem, as shown in (7) guarantees that its

output is always anti-monotonic with $\vec{\xi}$, which is a necessary condition any candidate solution must fulfil.

Note, however, that the approach presented in this section only works if the multivariate allocation problem (3) has an explicit solution $(Y_1(S), \dots, Y_d(S))$ that can then be used in the optimisation of (5) and its discretised version in (6a). We will show hereafter that a suitably designed algorithm makes it possible to conduct this optimisation with high precision, the basic reason being that the multivariate portfolio problem is reduced to a one-dimensional problem. When this approach cannot be used (that is, when the multivariate allocation problem (3) does not have an optimal solution), then it is still possible to design a slower numerical approach that can be generally applied. We develop this general approach in Section 5.

3.1. Algorithm

The algorithm we propose is motivated by the fact that a standard solver usually fails to solve a problem whenever the number of unknowns is too large; a fairly accurate initial guess becomes absolutely crucial to obtain accurate approximations (see Section 3.3). Therefore, to overcome this curse of dimensionality, we gradually increase the number of unknowns by feeding an initial guess to the algorithm that becomes more and more accurate.

The algorithm consists of two distinct components : the *optimisation*[‡] and the *interpolation*[§] iterated R times (also called runs). To obtain multivariate optimal solutions for the objective function, the procedure approximates optimal solutions $X_1^*, \dots, X_d^*$ by discretisation, attaining approximate optimal solutions $\tilde{X}_1^*, \dots, \tilde{X}_d^*$. It thus follows that the number of discretisation points used in each run (iteration) of the algorithm, denoted by n_r for $r = 0, 1, \dots, R$, must be selected for the procedure and the optimisation software. Starting from a fewer discretisation points ensures that the optimiser is able to solve the problem. Based on some numerical trials, we propose the following rule for determining the discrete points n_r for each run: $n_r = 2^r n_0$, $r = 0, 1, \dots, R$, with $n_0 = 20$.

To elucidate how the algorithm works, let us consider the case where $R = 2$ (i.e. three runs $r = 0, 1, 2$) for the variable ${}_r X^{(\cdot)}$.[¶]

Run: $r = 0$. The initial guess ${}_0 X^{(0)}$ is discretised with $n_0 = 20$ starting points that can be inserted into the optimisation software (orange circle of run $r = 0$ in

[‡] Note that the algorithm is implemented in MATLAB and the *optimisation* is carried out using the built-in function *fmincon*. Numerical experiments are conducted on a system featuring an Intel Core i7-8565U CPU @ 1.80GHz, 16GB of RAM and running Windows 11 Pro.

[§] In this context, the term *interpolation* is used to refer to both interpolation and extrapolation techniques. The linear method has been selected as the preferred approach, although alternative methodologies may be employed. See Bernard *et al.* (2019) for the pseudo-code and details about interpolation and extrapolation.

[¶] For sake of clarity, we indicate with ${}_r X^{(0)}$ the initial starting points of the optimisation (i.e. the right superscript is equal to zero) and with ${}_r X^{(M_r)}$ the solutions obtained from the optimisation (i.e. the right superscript is equal to M_r for $r = 0, 1, \dots, R$). The left subscript specifies the current run of the algorithm.

[†] See examples in Section 4 for the choice of $f(\cdot)$.

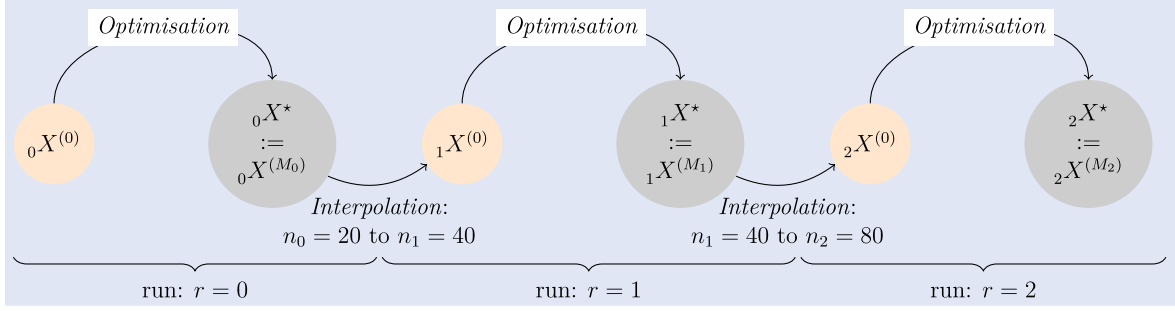


Figure 1. Diagram of the algorithm.

figure 1). Then the algorithm launches the *optimisation* that will operate until optimisation conditions are met at M_0 . Once satisfied, the output ${}_0X^{(M_0)}$, which is the best solution provided by the optimisation, is the numerical optimal solution ${}_0X^* := {}_0X^{(M_0)}$ of run $r = 0$ (grey circle of figure 1). Note that at this point, the numerical optimal solution ${}_0X^*$ approximates with $n_0 = 2^0 \cdot 20 = 20$ discretisation points the optimal solution X^* .

Run: $r = 1$. At this stage, the algorithm starts the run $r = 1$, by doubling the number of discretisation points $n_1 = 2^1 \cdot 20 = 40$ and using *interpolation*, expands the estimate ${}_0X^*$ to ${}_1X^{(0)}$ (second orange circle in figure 1). The variable ${}_1X^{(0)}$ corresponds to the new set of starting points that serve as input in the solver for the run $r = 1$. Next the *optimisation* is executed and the optimiser finds a new candidate solution ${}_1X^* := {}_1X^{(M_1)}$ (see grey circle of $r = 1$ in figure 1). At the run $r = 1$, the numerical optimal solution ${}_1X^*$ approximates the optimal solution X^* with a number of discretisation points equal to $n_1 = 40$.

Run: $r = 2$. Now, since $R = 2$, the algorithm computes the last run (i.e. $r = 2$). First, it doubles the discretisation points from $n_1 = 40$ to $n_2 = 2^2 \cdot 20 = 80$ and, using *interpolation*, it expands the variable ${}_1X^*$ to ${}_2X^{(0)}$, which is the new starting point (see orange circle in figure 1). The last *optimisation* is performed and the solver finds ${}_2X^* := {}_2X^{(M_2)}$. Note that, since this is the last run, the final numerical optimal solution of the routine is defined as $\tilde{X}^* := {}_2X^*$ and approximates the optimal solution X^* with $n_2 = 80$ discretisation points.

This whole procedure can be repeated R times. In the course of our numerical experiments (see Section 4.1 and Appendix 2), we have observed that, for each run, the algorithm fine-tunes the results and approaches the optimal one; the final numerical optimal solution, defined as $\tilde{X}^* := {}_RX^*$, is the one that most closely approximates the optimal solution X^* . Such a conclusion is reached because of the careful choice of the initial portfolio to fill in the optimiser. Starting with a fewer discretisation points ($n_0 = 20$) and feeding the optimiser at each iteration (run) with increasingly accurate initial guesses (achieved through the combination of *optimisation* and *interpolation* in the preceding run) ensures that the routine is capable of solving the problem.

As evidenced by our numerical experiments (see, for instance, Section 4.1 and Appendix 2), we observed that the

value of objective function $V(\cdot)$ tends to generally increase with each run (or as the number of points grows), followed by a stabilisation beyond a certain stage. This behaviour suggests a general pattern of improvement and a tendency towards convergence to the optimum in practice as the number of points increases. Nonetheless, it is important to discuss some possible potential limitations and conditions under which the algorithm may fail to converge. For instance, (i) number of starting initial guess. Feeding the solver with too many unknowns at the initial stage can complicate the optimisation process. (ii) Quality of the initial guess. A poor initial guess or a too coarse initial discretisation can lead the optimiser to poor outcomes, thereby limiting the algorithm to improve the value of the objective function $V(\cdot)$ in subsequent runs. (iii) Possibility to converge to a local optimum: the algorithm can reach a point where additional runs provide only minimal or no improvement, leading to early stabilisation before attaining the true optimum of the objective function $V(\cdot)$.

In Section 3.3, we illustrate a scenario where the value of the objective function $V(\cdot)$ begins to decrease as result of omitting the interpolation component from the algorithm combined with an initial guess that uses a large number of discretization points n .

3.2. Accuracy assessment of the numerical procedure

To assess the performance and accuracy of the MtU approach in solving multivariate optimal portfolio problems, we propose to analyse the error measure if an analytical solution exists (e.g. when the objective is a weighted sum of two exponential resp. two CRRA utility functions, see Section 4.1 resp. Appendix 2). In practice, we analyse for each iteration $r = 0, 1, \dots, R$ the discrepancy between the solution obtained numerically and the explicit solution by computing the mean relative absolute errors (MRAE), denoted by $EP_{r,k}^R$, $k = 1, \dots, d$, for the multivariate optimal portfolio solutions and by the relative absolute errors (RAE) for the objective function $V(\cdot)$, which we denote by EV_r^R , $r = 1, \dots, R$. Let $(\tilde{X}_1^*, \dots, \tilde{X}_d^*) := \tilde{x}_{jk}^*$, and $(X_1^*, \dots, X_d^*) := x_{jk}^*$, $j = 1, \dots, n$, $k = 1, \dots, d$ represent respectively the optimal solutions of the numerical approach and the optimal solutions of the analytical approach (obtained from simulating realisations of the analytically derived cdf). Then, the MRAE, for

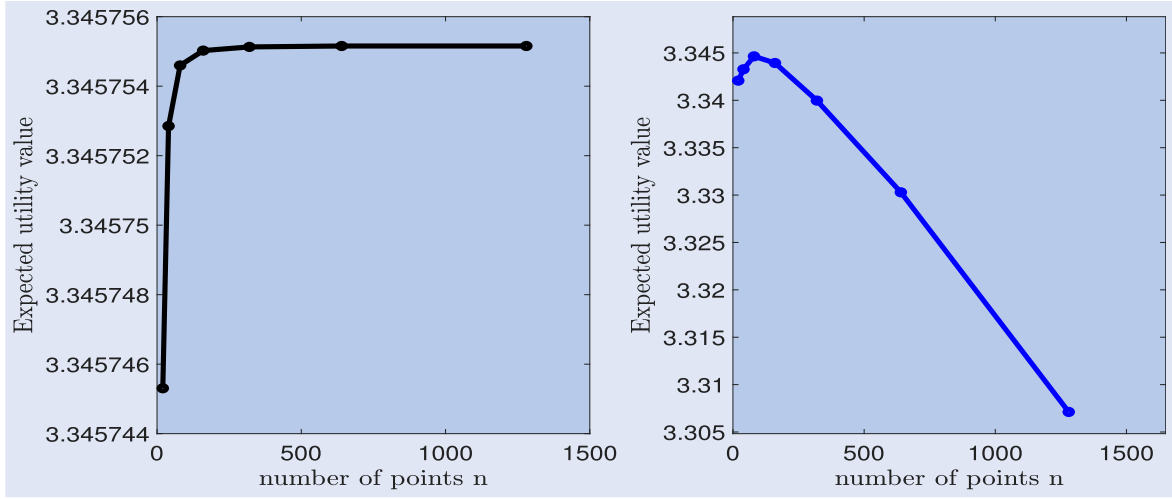


Figure 2. Example of optimisation failure without the *interpolation* part. The left panel shows the behaviour of expected utility when both *optimisation* and *interpolation* are applied (as outlined by the algorithm in Section 3.1). The right panel illustrates the behaviour of expected utility when only *optimisation* is used, without *interpolation* step.

$r = 0, 1, \dots, R$ and $k = 1, \dots, d$, is defined as

$$\text{EP}_{r,k}^R := \frac{1}{n} \sum_{j=1}^n \left| \frac{\tilde{x}_{jk}^* - x_{jk}^*}{x_{jk}^*} \right|.$$

Let $V(\tilde{X}_1^*, \dots, \tilde{X}_d^*)$ and $V(X_1^*, \dots, X_d^*)$ denote respectively the value of the objective function for the numerical approach and for the explicit solution. The relative absolute error, for $r = 0, 1, \dots, R$, is given by

$$\text{EV}_r^R := \left| \frac{V(\tilde{X}_1^*, \dots, \tilde{X}_d^*) - V(X_1^*, \dots, X_d^*)}{V(X_1^*, \dots, X_d^*)} \right|.$$

3.3. Example of optimisation failure

Hereafter, we report an example of the failure for an optimiser to solve a problem in the case that one inserts an initial guess with a number of discretisation points n large enough and without the *interpolation* part of the algorithm. For such example, we consider the case of a weighted sum of two CRRA utility functions in the case of homogeneous preferences for which we know the explicit form of the optimal portfolio (see Appendix 2 for more details).

The left panel of figure 2 displays the behaviour of the expected utility value when the number of discretisation points n increases using the algorithm discussed in Section 3.1 (i.e. the combination of *optimisation* and *interpolation*). Note that the value of expected utility increases and then remains equal to the previous one when the number of discretisation points grows up.

Instead, the right panel of figure 2 shows the behaviour of expected utility value using only the *optimisation* part. It can be observed that at runs $r = 0, 1, 2$ the optimiser is fed with a set of initial points (respectively 20, 40 and 80) for which the *optimisation* is able to yield good outcomes since the expected utility value is close to the one of the algorithm (left-hand panel) and increases. However, from run $r = 3$ onwards, the initial points provided to the optimiser become increasingly

large (respectively 160, 320, 640 and 1280), thereby rendering the *optimisation* a highly challenging endeavour and consequently producing poor outcomes. Indeed, the value of the expected utility starts to decrease when n becomes larger than $n = 80$ discretisation points. This is because the starting points provided to the solver are of poor quality, which is a direct consequence of the absence of fine-tuning through the *interpolation*.

4. Optimal multivariate expected utility with MtU

To verify the validity and the accuracy of the Multivariate-to-Univariate approach, we provide some numerical analyses if the multivariate objective function is given as a weighted sum of expected utilities[†] (used, for instance, in the context of collective assets allocation; see Alserda *et al.* 2019, Chen *et al.* 2021).

Let the multivariate law-invariant objective function $V(\cdot)$ be a multivariate expected utility given as

$$V(X_1, \dots, X_d) := \mathbb{E}[\mathcal{U}(X_1, \dots, X_d)] = \mathbb{E} \left[\sum_{i=1}^d k_i U_i(X_i) \right], \quad (8)$$

where $\mathcal{U}(\cdot)$ denotes a weighted sum of expected d utility functions $U_i(\cdot)$ for $i = 1, \dots, d$.

[†] As discussed in Balter and Schweizer (2024), a classical problem of utilitarian social preferences is the utility monster of Nozick (1974), an agent whose (marginal) utility for resources is so great that it dominates the planners' preferences. Furthermore, individual utilities are only identified up to addition and multiplication with arbitrarily large positive numbers. In contrast, certainty equivalents (see also Desmettre and Steffensen 2023 for further details) cannot be scaled arbitrarily. By translating agents' individual utilities into certainty equivalents, the planner converts them into the same monetary units before adding them up. This avoids the problems of the utility monster and of adding up incomparable quantities. However, our approach does not apply to the use of certainty equivalents. We thus recognise that this is a limitation of our work and that it applies on the wealth.

Let us consider a one-dimensional Black-Scholes market with drift μ , volatility σ and a constant, continuously compounded risk-free interest rate r ; i.e. $S_T = S_0 \exp\{(r - \frac{1}{2}\sigma^2)T + \sigma W_T\}$ where $W_T \sim \mathcal{N}(0, T)$. Let $\theta = (\mu - r)/\sigma$ (namely, in finance terms, the Sharpe ratio). The state-price process ξ_T is then unique and has a log-normal distribution with mean $\mu_{\xi_T} = (-r - \frac{1}{2}\theta^2)T$ and variance $\sigma_{\xi_T}^2 = \theta^2 T$; that is, $\xi_T \sim \log \mathcal{N}(\mu_{\xi_T}, \sigma_{\xi_T}^2)$. Specifically,

$$\xi_T = \xi_0 \exp\left\{\left(-r - \frac{1}{2}\theta^2\right)T - \theta W_T\right\} \quad (9)$$

where we assume, without loss of generality, $\xi_0 = 1$.

When the objective function $V(\cdot)$ is defined as in (8), analytical solutions to problem (1) are available in some special case. Proposition 4.1 summarises the solution present in the literature (see e.g. Bernard *et al.* 2023, Proposition B.1) and expands it to the general case of weighted sum of expected utilities.

PROPOSITION 4.1 *Consider the portfolio optimisation problem (1) in which $V(x_1, \dots, x_d) = \mathbb{E}[\sum_{i=1}^d k_i U_i(x_i)]$ such that, for $i = 1, \dots, d$, U_i is concave and continuously differentiable over a subset of $\mathbb{R}$ and U_i' is invertible at any point in which U_i is finite (otherwise, we assume that $U_i(x_i)$ is $-\infty$) and $k_i > 0$. Then, the optimal solution $(X_i^*)_{i=1, \dots, d}$ to problem (1) is given by*

$$X_i^* := [k_i U_i']^{-1}(\lambda_i^* \xi_T), \quad i = 1, \dots, d, \quad (10)$$

where $\lambda_i^* > 0$ are such that $(\lambda_1^*, \dots, \lambda_d^*)$ solves

$$\begin{aligned} \max_{(\lambda_1, \lambda_2, \dots, \lambda_d) \in \Lambda} \mathbb{E} \left[\mathcal{U} \left([k_1 U_1']^{-1}(\lambda_1 \xi_T), \right. \right. \\ \left. \left. \dots, [k_d U_d']^{-1}(\lambda_d \xi_T) \right) \right], \end{aligned} \quad (11)$$

in which $\Lambda = \{(\lambda_1, \lambda_2, \dots, \lambda_d) \in \mathbb{R}_+^d \mid \mathbb{E}[\xi_T \sum_{i=1}^d [k_i U_i']^{-1}(\lambda_i \xi_T)] = w_0\}$.

Proof See Appendix A.1. ■

Consider d agents (e.g. banks) and assume that each agent i with utility function $U_i(x)$ holds a random variable (e.g. risk) at a certain point of time. We define the total risk of all agents as S .

A semi-explicit optimal risk sharing can be found when the objective function $V(\cdot)$ is given by (8) by solving a system of equations given in the following proposition, which is based on the theorem of Borch (1962) and some arguments of Gerber and Pafumi (1998).

PROPOSITION 4.2 *When utility functions are differentiable, the risk-sharing problem (3) is maximised by optimal allocations $(\check{X}_1, \dots, \check{X}_d)$ if and only if*

$$\begin{aligned} \forall i \in \{1, \dots, d\}, \\ \forall j \in \{1, \dots, d\} \quad k_i U_i'(\check{X}_i) = k_j U_j'(\check{X}_j) \quad \text{a.s.} \end{aligned} \quad (12)$$

where $U_i'(\cdot)$ is the so-called marginal utility (i.e. the first derivative with respect to the i -th argument).

Proof See Theorem 1 of Gerber and Pafumi (1998, Section 8, p. 80). ■

We apply the Multivariate-to-Univariate approach to exponential utility functions. The numerical outputs are compared with the analytical solutions derived from Proposition 4.1. We reproduce the study with CRRA utility functions in Appendix 2.

4.1. The case of weighted sum of exponential utility functions

In this section, we study the special case of a weighted sum of two and six exponential utility functions. Specifically, the collective utility function $\mathcal{U}$ is written as

$$\mathcal{U}(X_1, \dots, X_d) = \sum_{i=1}^d k_i U_{\gamma_i}(X_i) \quad (13)$$

where $U_{\gamma_i}(X_i)$ is a univariate utility function of the following form

$$U_{\gamma_i}(x) = -e^{-\gamma_i x}, \quad i = 1, \dots, d. \quad (14)$$

The parameter $k_i > 0$ is the weight of the utility function $\mathcal{U}$ and the parameter $\gamma_i > 0$, $i = 1, \dots, d$, is the risk aversion coefficient of the exponential utility function.

We derive analytical solutions in the case of a weighted sum of exponential utility functions (see Propositions 4.3 and 4.4) and compare them with the numerical approach, when the pricing kernel is defined as in (9).

PROPOSITION 4.3 *The optimal solution (X_1^*, X_2^*) to the problem*

$$\max_{(X_1, X_2) \in \mathcal{A}} \mathbb{E} \left[k_1 (-e^{-\gamma_1 X_1}) + k_2 (-e^{-\gamma_2 X_2}) \right], \quad (15)$$

in which $\mathcal{A} := \{(X_1, X_2) : \mathbb{E}[\xi_T(X_1 + X_2)] = w_0\}$ is given by

$$\begin{pmatrix} X_1^* \\ X_2^* \end{pmatrix} = \begin{pmatrix} w_0 \lambda^* e^{rT} - \frac{1}{\gamma_1} \left(rT - \frac{\theta^2 T}{2} \right) - \frac{\ln(\xi_T)}{\gamma_1} \\ w_0 (1 - \lambda^*) e^{rT} - \frac{1}{\gamma_2} \left(rT - \frac{\theta^2 T}{2} \right) - \frac{\ln(\xi_T)}{\gamma_2} \end{pmatrix} \quad (16)$$

where $\lambda^* = \frac{\ln(\frac{k_1 \gamma_1}{k_2 \gamma_2}) + \gamma_2 w_0 e^{rT}}{(\gamma_1 + \gamma_2) w_0 e^{rT}}$.

Proof See Appendix A.2. ■

Next, Proposition 4.4 shows the explicit solution of problem (1) in the case of a weighted sum of d exponential utility functions.

PROPOSITION 4.4 *The optimal solution $(X_1^*, X_2^*, \dots, X_d^*)$ to the problem*

$$\max_{(X_1, X_2, \dots, X_d) \in \mathcal{A}} \mathbb{E} \left[\sum_{i=1}^d k_i (-e^{-\gamma_i X_i}) \right], \quad (17)$$

in which $\mathcal{A} := \{(X_1, X_2, \dots, X_d) : \mathbb{E}[\xi_T \sum_{i=1}^d X_i] = w_0\}$ is given by

$$X_i^* = w_0 \lambda_i^* e^{rT} - \frac{1}{\gamma_i} \left(rT - \frac{\theta^2 T}{2} \right) - \frac{\ln(\xi_T)}{\gamma_i}, \quad i = 1, \dots, d, \quad (18)$$

where the parameters $\lambda_i^*, i = 1, \dots, d$, are

$$\lambda_i^* = \frac{\gamma_d}{\gamma_i} \lambda_d^* - \frac{1}{\gamma_i w_0 e^{rT}} \ln \left(\frac{k_d \gamma_d}{k_i \gamma_i} \right) \quad i = 1, \dots, d-1,$$

$$\lambda_d^* = \frac{1 + \sum_{i=1}^{d-1} \frac{1}{\gamma_i w_0 e^{rT}} \ln \left(\frac{k_d \gamma_d}{k_i \gamma_i} \right)}{1 + \gamma_d \sum_{i=1}^{d-1} \frac{1}{\gamma_i}}.$$

Proof See Appendix A.3. ■

We present the numerical analysis of the MtU method, which is compared with the explicit expressions from Propositions 4.3 and 4.4 to study their respective accuracy.

To implement the MtU method, we need the following proposition. Proposition 4.5 shows the optimal risk sharing for d dimensions using Proposition 4.2.

PROPOSITION 4.5 *Let $\frac{1}{\gamma} = \frac{1}{\gamma_1} + \dots + \frac{1}{\gamma_d}$. In the case of exponential utility functions, the optimal risk sharing $(X_1, X_2, \dots, X_d)$ of S is given, for each $i = 1, \dots, d$, by*

$$X_i := Y_i(S) = \frac{\gamma}{\gamma_i} S + \frac{\ln(k_i \gamma_i)}{\gamma_i} - \frac{\gamma}{\gamma_i} \sum_{j=1}^d \frac{\ln(k_j \gamma_j)}{\gamma_j}. \quad (19)$$

Proof See Appendix A.4. ■

We now use this proposition to implement the MtU method respectively when the dimension is $d = 2$ (Section 4.1.1) and $d = 6$ (Section 4.1.2). The input parameters are displayed directly in the figure captions. The weights $k_i, i = 1, \dots, d$, are chosen arbitrarily and the global budget is $w_0 = 10$.

4.1.1. The weighted sum of two exponential utility functions. Given the explicit solutions of Proposition 4.5 and the quantile formulation (see Proposition 2.1), then the optimal portfolio choice problem (1) is written as

$$\sup_{F_S^{-1} \text{ s.t. } \mathbb{E}[\xi_T F_S^{-1}(H_T)] = w_0} \mathbb{E} \left[k_1 \left(-e^{-\gamma_1 Y_1(F_S^{-1}(H_T))} \right) + k_2 \left(-e^{-\gamma_2 Y_2(F_S^{-1}(H_T))} \right) \right] \quad (20)$$

where $Y_1(\cdot)$ and $Y_2(\cdot)$ are defined in (19). Only in this example, we repeat explicitly the problem for clarifying how the method works, but the same idea applies for the cases of $d = 6$ (Section 4.1.2) and for the CRRA framework too (Appendix 2).

Applying the algorithm for the MtU method with a number of runs $R = 5$ leads to the outcome of figure 3. The picture exhibits the numerical optimal payoffs[†] $\tilde{X}_1^*$ and $\tilde{X}_2^*$, denoted by the marker symbol *, with $n_5 = 20 \cdot 2^5 = 640$ discretisation points and the optimal payoffs X_1^* (solid blue line) and X_2^* (solid red line) computed through Proposition 4.3. The MtU approach and the analytical solution provide a close match,

[†] For sake of clarity, we recall that the numerical optimal payoff is denoted by $\tilde{X}^*$ and defined as $\tilde{X}^* := {}_R X^*$. Given that $d = 2$ and $R = 5$, we have $\tilde{X}_1^* := {}_5 X_1^*$ and $\tilde{X}_2^* := {}_5 X_2^*$.

indicating that such a numerical method works well in approximating the optimal payoffs of a weighted sum of exponential utility functions. Such approach is well-suited if the optimal risk sharing is known (Proposition 4.5).

To assess the accuracy of the MtU method, figure 4 provides the mean relative absolute error, as illustrated in Section 3.2, between the numerical and analytical solutions for payoffs X_1^* and X_2^* .

As shown in figure 4, the two error measures MRAE for X_1^* and X_2^* both decrease quickly when the number of discretisation points increases, meaning that there is a fast convergence of numerical optimal solutions to the exact ones. The results of figure 4 validate the performance of the MtU approach. In addition to that, the left panel of figure 5 displays the behaviour of the expected utility value of the numerical optimal solutions as the number of discretisation points increases in the case of a weighted sum of exponential utility functions. While the right panel of figure 5 exhibits the relative absolute error between the utility value computed by the numerical approach and the one obtained by means of the analytical expression. Both graphs corroborate, once again, the validity of the numerical approach in approximating the optimal payoffs.

4.1.2. The weighted sum of six exponential utility functions. We now extend the analysis to six payoffs, X_i for $i = 1, \dots, 6$. This leads to a multivariate portfolio choice problem where the expected utility is characterised as a weighted sum of six exponential utility functions. Figure 6 exhibits the numerical optimal solutions obtained using the MtU approach for the payoffs $\tilde{X}_i^*, i = 1, \dots, 6$, with $n_5 = 640$ discretisation points, indicated by the symbol *. The optimal payoffs derived via Proposition 4.4 are represented by solid lines. As displayed, the outcomes obtained with the MtU approach overlap the optimal payoffs.[‡]

Figure 7 presents the mean relative absolute error for each payoff, highlighting its reduction as the number of discretisation points increases. Figure 8 illustrates the behaviour of both the expected utility value and the relative absolute error as functions of discretisation granularity.

4.1.3. Computation speed and accuracy of the MtU approach. As follows, we assess the performance of the Multivariate-to-Univariate approach as the dimensionality d of the problem increases, focusing on both accuracy and computational speed. The accuracy is evaluated using the relative absolute error (RAE) between the utility value computed numerically and the corresponding valued obtained by means of the analytical expression. Such a comparison is carried out under a discretisation setting of $n_5 = 20 \cdot 2^5 = 640$, which corresponds to $R = 5$. The computational speed is measured in seconds and reflects the total time required to complete a number of runs $R = 5$.

As an illustrative example, we consider the case in which $d \in \{2, 3, 6, 9, 12\}$. In each case, we assign equal weights

[‡] Additional experiments, such as a 12-dimensional example, have been also run and are available upon request.

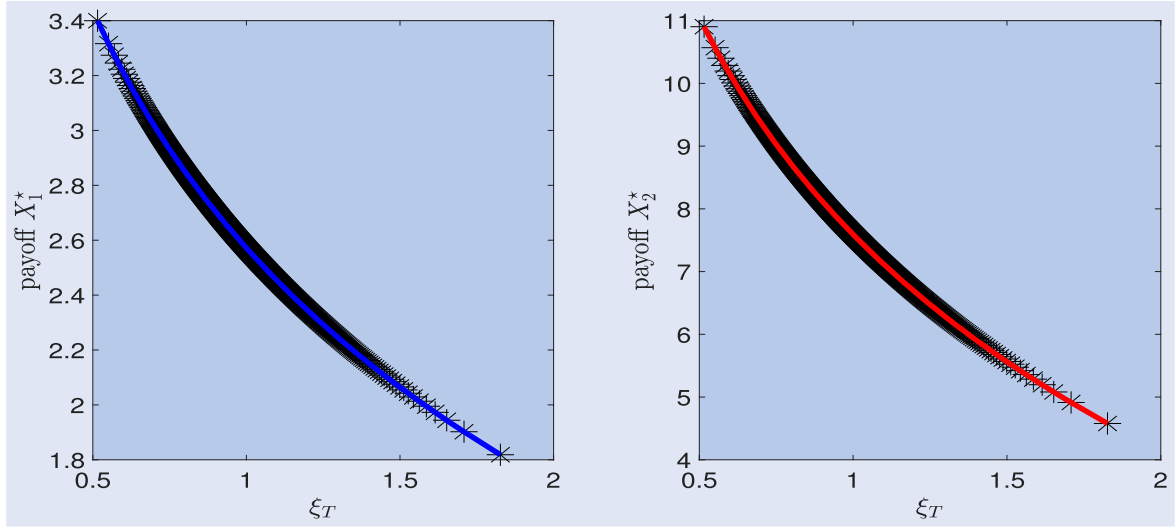


Figure 3. Optimal portfolios in the case of a weighted sum of two exponential utility functions using the MtU approach. Input parameters: $\gamma_1 = 0.8$, $\gamma_2 = 0.2$, $k_1 = 0.3$, $k_2 = 0.7$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$, $T = 1$, and $n_5 = 640$.

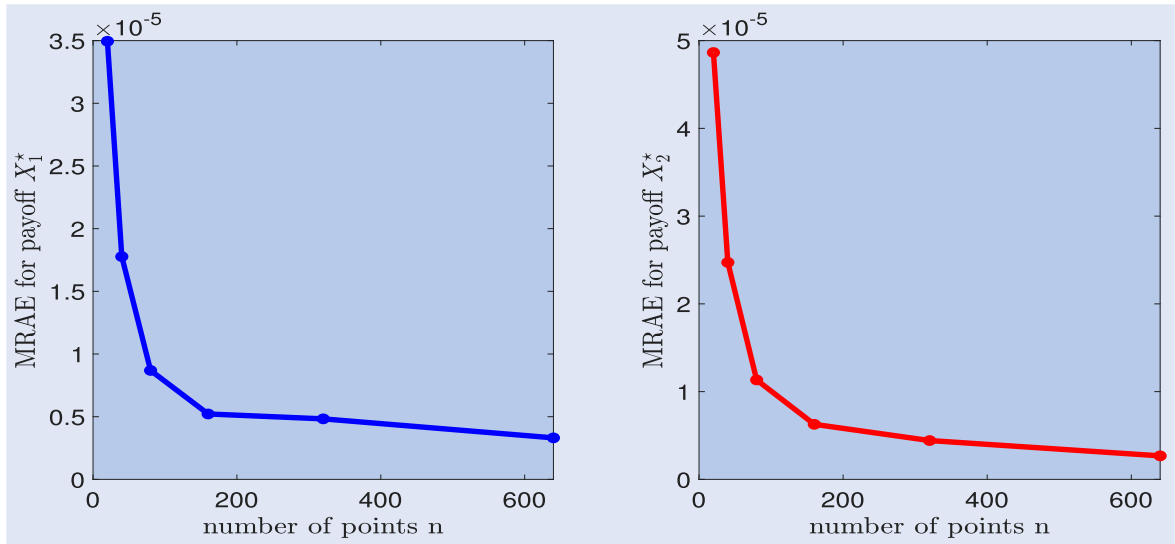


Figure 4. Mean relative absolute error in the case of a weighted sum of two exponential utility functions using the MtU approach. Input parameters: $\gamma_1 = 0.8$, $\gamma_2 = 0.2$, $k_1 = 0.3$, $k_2 = 0.7$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$ and $T = 1$.

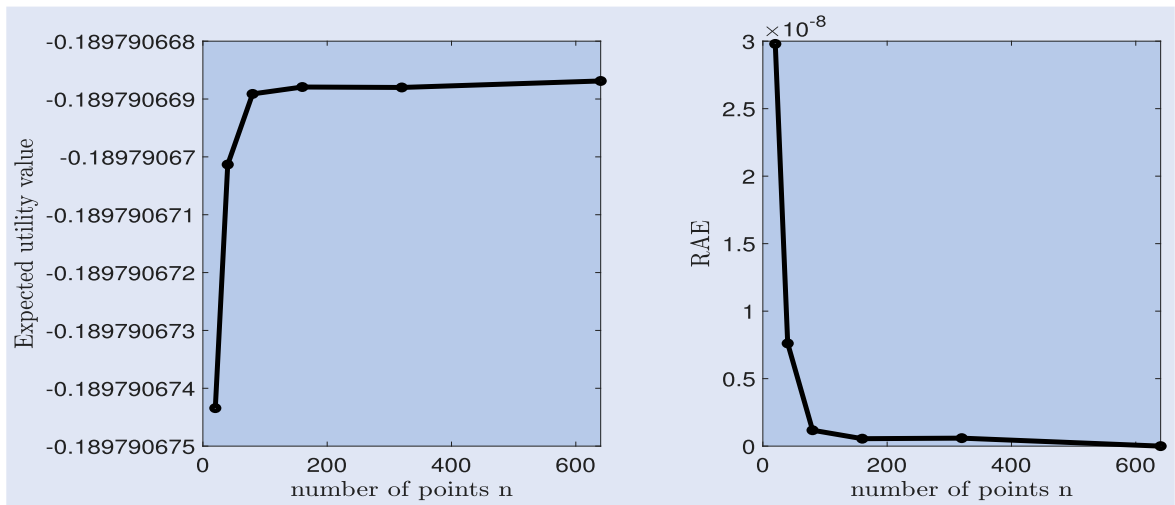


Figure 5. Expected utility value and relative absolute error in the case of a weighted sum of two exponential utility functions using the MtU approach. Input parameters: $\gamma_1 = 0.8$, $\gamma_2 = 0.2$, $k_1 = 0.3$, $k_2 = 0.7$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$, and $T = 1$.

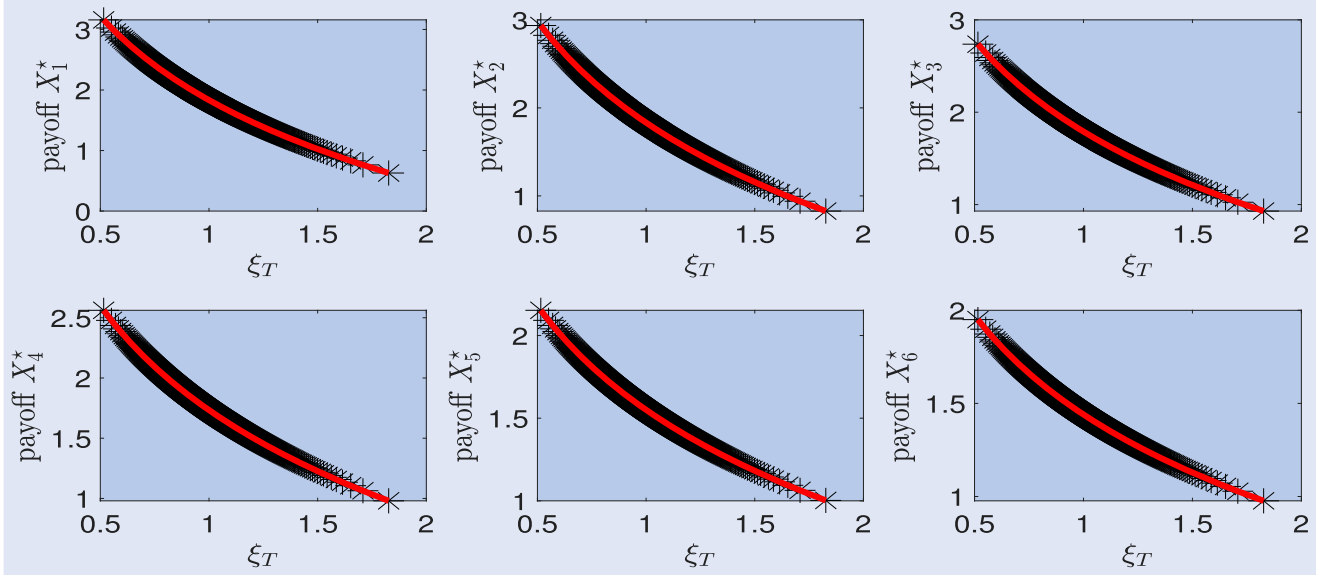


Figure 6. Optimal portfolios in the case of a weighted sum of six exponential utility functions using the MtU approach. Input parameters: $\gamma_1 = 0.5, \gamma_2 = 0.6, \gamma_3 = 0.7, \gamma_4 = 0.8, \gamma_5 = 1.1, \gamma_6 = 1.3, k_1 = k_2 = k_3 = k_4 = k_5 = k_6 = 1/6, \mu = 0.05, r = 0.01, \sigma = 0.2, T = 1$, and $n_5 = 640$.

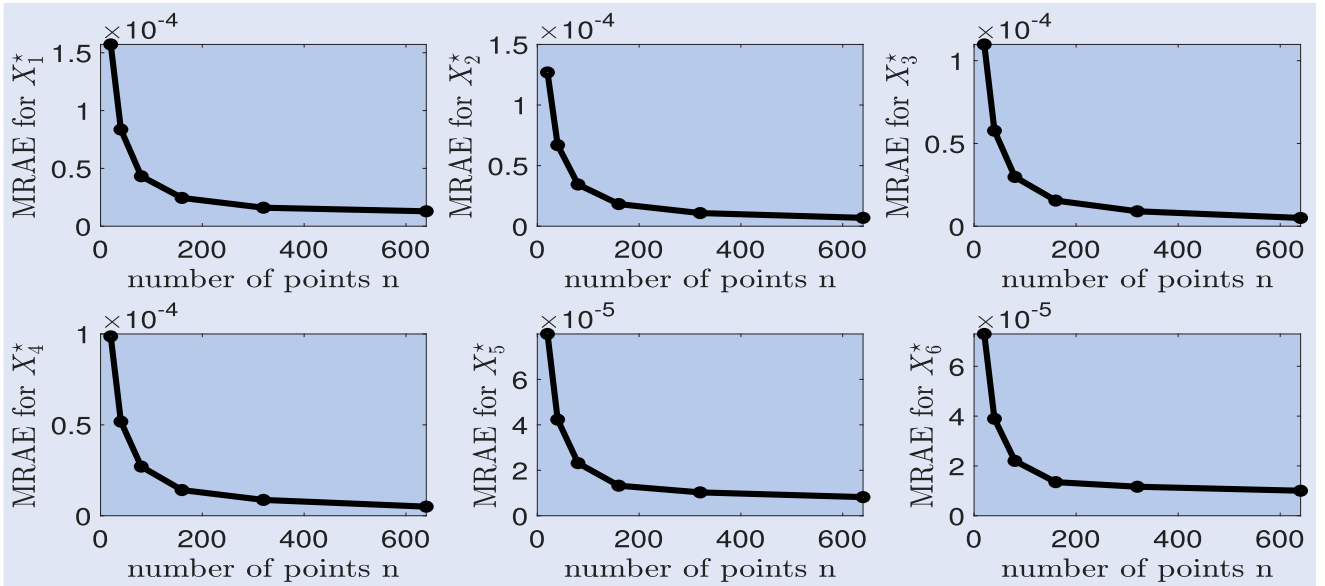


Figure 7. Mean relative absolute error in the case of a weighted sum of six exponential utility functions using the MtU approach. Input parameters: $\gamma_1 = 0.5, \gamma_2 = 0.6, \gamma_3 = 0.7, \gamma_4 = 0.8, \gamma_5 = 1.1, \gamma_6 = 1.3, k_1 = k_2 = k_3 = k_4 = k_5 = k_6 = 1/6, \mu = 0.05, r = 0.01, \sigma = 0.2$ and $T = 1$.

$k_i = 1/d$ for $i = 1, \dots, d$ and the specific configurations of individual risk aversion coefficients are as follows:

Case $d = 2$: $\gamma_1 = 0.5$ and $\gamma_2 = 0.7$,

Case $d = 3$: $\gamma_1 = 0.5, \gamma_2 = 0.7$, and $\gamma_3 = 1.3$;

Case $d = 6$: $\gamma_1 = 0.5, \gamma_2 = 0.6, \gamma_3 = 0.7, \gamma_4 = 0.8, \gamma_5 = 1.1$, and $\gamma_6 = 1.3$;

Case $d = 9$: $\gamma_1 = 0.5, \gamma_2 = 0.6, \gamma_3 = 0.7, \gamma_4 = 0.8, \gamma_5 = 0.9, \gamma_6 = 1.0, \gamma_7 = 1.1, \gamma_8 = 1.2$, and $\gamma_9 = 1.3$;

Case $d = 12$: $\gamma_1 = 0.8, \gamma_2 = 0.9, \gamma_3 = 1.0, \gamma_4 = 1.1, \gamma_5 = 1.2, \gamma_6 = 1.3, \gamma_7 = 1.4, \gamma_8 = 1.5, \gamma_9 = 1.6, \gamma_{10} = 1.7, \gamma_{11} = 1.8$, and $\gamma_{12} = 1.9$.

The remaining input parameters are common across all cases and are the following ones: $\mu = 0.05, r = 0.01, \sigma = 0.2$, and $T = 1$.

In figure 9, we observe that although the relative absolute error (blue line) rises with the dimension d , it remains relatively low across all dimensions, on the order of 10^{-9} . This suggests that the Multivariate-to-Univariate approach remains robust as the dimensionality d grows (at least up to $d = 12$). The computation time (red line) increases steadily with the dimension d , ranging from 300 s at $d = 2$ to about 650 s at $d = 12$.

The following section proposes a crude method, named the General Numerical (also labelled as GNum) approach

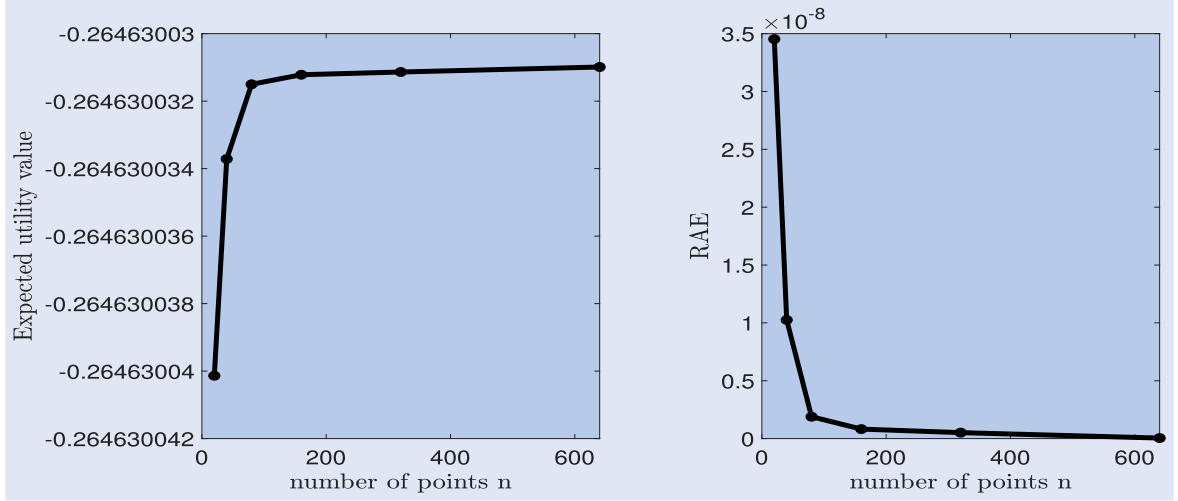


Figure 8. Expected utility value and relative absolute error in the case of a weighted sum of six exponential utility functions using the MtU approach. Input parameters: $\gamma_1 = 0.5$, $\gamma_2 = 0.6$, $\gamma_3 = 0.7$, $\gamma_4 = 0.8$, $\gamma_5 = 1.1$, $\gamma_6 = 1.3$, $k_1 = k_2 = k_3 = k_4 = k_5 = k_6 = 1/6$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$, and $T = 1$.

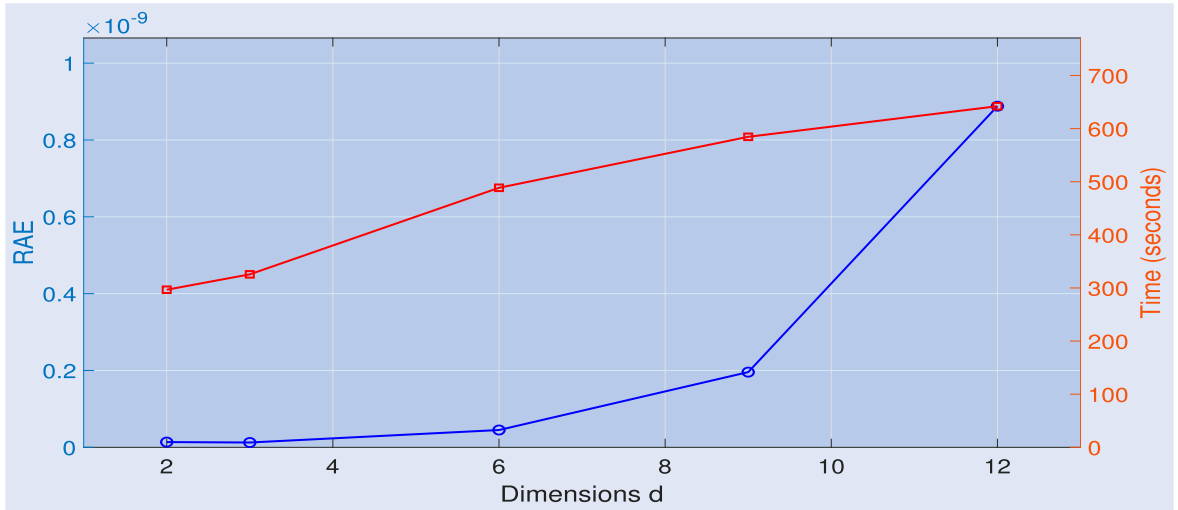


Figure 9. Effect of dimensionality d on relative absolute error (blue line, left y-axis) and computation time measured in seconds (red line, right y-axis).

that can be employed as an alternative to the Multivariate-to-Univariate approach when the latter is inapplicable (in the absence of explicit or semi-explicit solutions to (3)). This alternative approach can be employed to address the multivariate portfolio selection problem when the dimensionality of the problem is small to moderate.

5. General numerical (GNum) approach

In general, there is no semi-closed-form expression of the risk-sharing problem (3) and thus it is not possible to reduce it to a one-dimensional optimisation problem (such as done in the MtU approach). We show how the problem (1) can be solved numerically exploiting the multivariate cost-efficiency (see Bernard *et al.* 2023) of the optimal solution. To this end, we first discretise in a similar way as in Section 3 the multivariate portfolio choice problem. Thus, we first discretise

the state-price process ξ_T so that we obtain n ordered distinct equiprobable states of the world $\xi_1 > \xi_2 > \dots > \xi_n$ (see Section 3).

Then we formalise the optimisation problem (1) for the appropriate choice of $f(\cdot)$ within a discrete setting

$$\sup_{(\vec{X}_1, \vec{X}_2, \dots, \vec{X}_d) \in \mathcal{A}'} f(\vec{X}_1, \vec{X}_2, \dots, \vec{X}_d), \quad (21a)$$

where $\mathcal{A}'$ corresponds to the discretised version of the admissible set $\mathcal{A}$. Specifically,

$$\mathcal{A}' = \left\{ (\vec{X}_1, \vec{X}_2, \dots, \vec{X}_d) \in (\mathbb{R}^n)^d \left| \sum_{i=1}^n \frac{1}{n} [\xi_i(x_{i1} + \dots + x_{id})] = w_0 \right. \right\}, \quad (21b)$$

where $\vec{X}_k = (x_{ik})_{i=1}^n$ for $k = 1, 2, \dots, d$. We can reduce the set in which we search for the solution by reformulating (21)

as follows. Let us define $\vec{S}$ as $\vec{S} = \vec{X}_1 + \vec{X}_2 + \dots + \vec{X}_d$ and replace the last variable $\vec{X}_d$ by $\vec{S} - \sum_{k=1}^{d-1} \vec{X}_k$. Then (21a) can be rewritten as

$$\sup_{(\vec{X}_1, \vec{X}_2, \dots, \vec{X}_{d-1}, \vec{S}) \in \mathcal{A}'} f\left(\vec{X}_1, \vec{X}_2, \dots, \vec{X}_{d-1}, \vec{S} - \sum_{k=1}^{d-1} \vec{X}_k\right), \quad (22a)$$

and (21b) as

$$\mathcal{A}' = \left\{ (\vec{X}_1, \vec{X}_2, \dots, \vec{X}_{d-1}, \vec{S}) \in (\mathbb{R}^n)^d \left| \sum_{i=1}^n \frac{1}{n} [\xi_i s_i] = w_0 \right. \right\}, \quad (22b)$$

where $\vec{S} = (s_i)_{i=1}^n$ with $s_i := \sum_{k=1}^d x_{ik}$ for $i = 1, 2, \dots, n$.

We look for n values of the d -dimensional vector $(\vec{X}_1, \vec{X}_2, \dots, \vec{X}_{d-1}, \vec{S})$ in which the budget constraint only constrains $\vec{S}$.

We can further restrict the set in which to look for an optimal solution by using the fact that the optimal solution to (1) is such that $\vec{\xi}$ and $\vec{S}$ are anti-monotonic. We then have that

$$\begin{aligned} \xi_1 &> \xi_2 > \dots > \xi_n \\ s_1 &\leq s_2 \leq \dots \leq s_n. \end{aligned}$$

Now let us define $\vec{Z} := (z_i)_{i=1}^n$, $z_i = s_i - s_{i-1}$, with the convention that $s_0 = 0$. Imposing that s_i , $i = 1, \dots, n$, are an increasing sequence it holds that

$$z_1 \leq z_1 + z_2 \leq \dots \leq z_1 + z_2 + \dots + z_n = s_n,$$

where the increasing constraint becomes simply $z_i \geq 0$ for all $i = 1, 2, \dots, n$.

The optimisation problem (22a) can be rewritten according to the appropriate choice of $f(\cdot)$ that depends on $V(\cdot)$; an example of such a problem is given in (23) for the expected utility setting and in (29) for the distorted expectation setting.

Solving the above optimisation problem requires the use of a solver for $d \times n$ dimensions. Indeed, there is a substantial difference between the MtU approach (Section 3) and the GNum method in terms of discretisation and resolution of the problem the solver deals with. Although the discretisation of ξ_T coincides for both numerical approaches, the starting points, or initial guess, to input in the solver differ between the two situations. For the Multivariate-to-Univariate approach, the number of starting points, and consequently the number of unknowns to determine, is equal to n_r (i.e. $x_1, x_2, \dots, x_{n_r}$), as discussed in Section 3, whereas for the General Numerical approach, the starting points, and consequently the unknowns, are $n_r \times d$ (i.e. $x_{1k}, x_{2k}, \dots, x_{n_r k}$ for $k = 1, \dots, d$). The functioning of the algorithm and the assessment of numerical accuracy are, respectively, founded upon the content of Sections 3.1 and 3.2.

5.1. The case of weighted sum of exponential utility functions

In this section, we implement the General Numerical approach in the case of an expected utility setting. In this case, (22) can be reformulated by the appropriate choice

of $f(\cdot)$ that depends on $V(\cdot)$ and the resulting optimisation problem (22a) becomes

$$\sup_{(x_{i1}, x_{i2}, \dots, x_{i(d-1)}, z_i)_{i=1, \dots, n} \in \mathcal{A}'} \frac{1}{n} \sum_{i=1}^n U\left(x_{i1}, x_{i2}, \dots, x_{i(d-1)}, \sum_{\ell=1}^i z_\ell - \sum_{k=1}^{d-1} x_{ik}\right), \quad (23a)$$

where the admissible set (22b) is given by

$$\mathcal{A}' = \left\{ (x_{i1}, x_{i2}, \dots, x_{i(d-1)}, z_i) \in \mathbb{R}^{d-1} \times (\mathbb{R}_+), \right. \\ \left. i = 1, 2, \dots, n \left| \sum_{i=1}^n \zeta_i z_i = w_0 \right. \right\} \quad (23b)$$

and $\zeta_i = \frac{1}{n} \sum_{k=i}^n \xi_k$, for $i = 1, \dots, n$.

To verify the validity and the accuracy of the aforementioned approach, we present the numerical analysis for a weighted sum of two resp. six exponential utility functions (Section 5.1.1 resp. Section 5.1.2) in which the analytical solution is available (Proposition 4.3 resp. Proposition 4.4) and used as a benchmark.

5.1.1. Weighted sum of two exponential utility functions.

We implement the GNum method with a sum of heterogeneous utility functions in which the starting points of payoffs X_1 and X_2 are different and it is not possible to know a priori how to allocate the share of budget to X_1 and X_2 . Thus, to demonstrate that the General Numerical approach is capable of approximating optimal payoffs by retrieving the numerical optimal optimal budget split between X_1 and X_2 , we proceed as follows.

We arbitrarily assume that $w_{0,1}$ and $w_{0,2}$, which are the allocated budget at time zero of respectively X_1 and X_2 , are equal to $w_0/2$. The optimal budget split, denoted by $w_{0,1}^*$ and $w_{0,2}^*$, can be obtained from the optimal solution (X_1^*, X_2^*) as: $w_{0,1}^* = \mathbb{E}[\xi_T X_1^*] = 2.5336$ and $w_{0,2}^* = \mathbb{E}[\xi_T X_2^*] = 7.4664$, satisfying $w_{0,1}^* + w_{0,2}^* = w_0$. These values provide a benchmark to show how the GNum method recovers the numerical optimal budget split between X_1 and X_2 at each run. To find the best approximation of the budget split between X_1 and X_2 , we compute a first cycle of the algorithm with a number of runs $R = 3$.

Run: $r = 0$. At the end of run $r = 0$ (see figure 1) of the first cycle in which the numerical optimal solutions ${}_0X_1^*$ and ${}_0X_2^*$ approximate with $n_0 = 20$ discretisation points the optimal solutions X_1^* and X_2^* , the numerical optimal budget split is given by ${}_0w_{0,1}^* = 2.5330$ and ${}_0w_{0,2}^* = 7.4670$.[†] This represents a significant adjustment from the initial 50-50 budget split and the updated numerical optimal budget split is then used as the basis for the beginning of run $r = 1$.

[†] We recall that the left subscript specifies the current run of the algorithm.

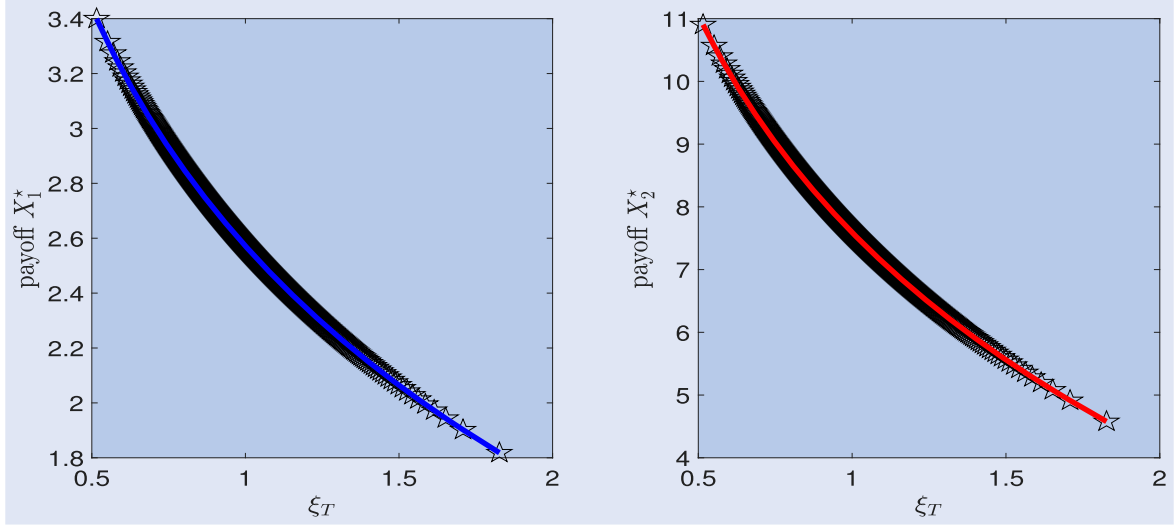


Figure 10. Optimal portfolios in the case of a weighted sum of two exponential utility functions using the GNum method. Input parameters: $\gamma_1 = 0.8$, $\gamma_2 = 0.2$, $k_1 = 0.3$, $k_2 = 0.7$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$, $T = 1$ and $n_5 = 640$.

Run: $r = 1$. At the end of run $r = 1$, where the numerical optimal solutions ${}_1X_1^*$ and ${}_1X_2^*$ approximate the optimal solution X_1^* and X_2^* using $n_0 = 40$ discretisation points, the numerical optimal budget split is updated to ${}_1w_{0,1}^* = 2.5333$ and ${}_1w_{0,2}^* = 7.4667$. We observe that the numerical optimal budget split is getting closer to the optimal budget split $w_{0,1}^*$ and $w_{0,2}^*$. These values are used to initialise run $r = 2$.

Run: $r = 2$. At the end of run $r = 2$, the numerical optimal solutions ${}_2X_1^*$ and ${}_2X_2^*$ approximate the optimal solutions X_1^* and X_2^* using $n_0 = 80$ discretisation points. The numerical optimal budget split is further refined to ${}_2w_{0,1}^* = 2.5335$ and ${}_2w_{0,2}^* = 7.4665$, which are used for the last run $r = 3$ of the first cycle.

Run: $r = 3$. Since $R = 3$, the algorithm computes the last run (i.e. run $r = 3$). At the end of this run, the numerical optimal solutions ${}_3X_1^*$ and ${}_3X_2^*$ approximate the optimal solutions X_1^* and X_2^* using $n_0 = 160$ discretisation points. The numerical optimal budget split is respectively ${}_3w_{0,1}^* = 2.5335$ and ${}_3w_{0,2}^* = 7.4665$, remaining stable up to the fourth decimal place.

At each run, the algorithm produces and updates the numerical budget split, which is progressively fine-tuned and then used as basis for the subsequent run.

Once a more fine-tuned budget split is obtained, it is possible to restart the algorithm (e.g. a second cycle) with a more accurate share of budget for X_1 and X_2 as input and the desired number of runs R (in our case, $R = 5$). By starting with a more accurate budget split, the algorithm is better positioned, already from the first run of the second cycle (run $r = 0$), to further fine-tune the budget split and, as a result, to provide better estimates in subsequent runs. Indeed, after re-optimising with the second cycle, we obtain numerical optimal solutions, denoted by the pentagram marker symbol in figure 10, which match the analytical ones, which are

X_1^* (solid blue line) and X_2^* (solid red line).[†] As figure 10 displays, the numerical optimal payoffs $\tilde{X}_1^*$ and $\tilde{X}_2^*$ obtained with the second cycle and $n_5 = 20 \cdot 2^5 = 640$ discretisation points overlap the exact ones, indicating that the GNum approach is also able to approximate the optimal payoffs of a weighted sum of exponential utility functions. As done in Section 4.1.1 for assessing the performance of the method, figure 11 provides the mean relative absolute error for X_1^* and X_2^* , while figure 12 shows respectively the expected utility value (left panel) and its relative absolute error (right panel) when the discretisation points increase. Both graphs validate the numerical approach developed in the beginning of this section.

For comparison, we now show the same example using $R = 5$ runs, starting from an initial 50-50 budget split (i.e. without inserting the fine-tuned budget split obtained from a first cycle). In this case, no second cycle is performed. From figure 13, we can observe that the initial mean relative absolute errors for X_1^* and X_2^* are relatively high, but then they start to decrease rapidly as the number of discretisation points increases. This is also due to the refinement of the budget split at each run, as previously discussed. Similarly, the relative absolute error of the expected utility value (see figure 14) is initially high but then exhibits a decreasing behaviour.

Each method, either a single cycle with $R = 5$ runs and starting 50-50 initial budget split or two cycles starting with the refined budget split, performs well and provides reliable results. The main difference lies in the magnitude of the mean relative absolute errors for X_1^* and X_2^* with $n_5 = 640$ discretisation points (order of 10^{-4} vs 10^{-6}), as well as the relative absolute error of the expected utility value with $n_5 = 640$ discretisation points (order 10^{-8} vs 10^{-10}).

5.1.2. Weighted sum of six exponential utility functions.

[†] Note that knowing the right share of budget for X_1 and X_2 is a crucial step in the procedure. Indeed, it is necessary to set the prices to the right share of budget for the optimisation and for the interpolation. In the Multivariate-to-Univariate approach, the share of budget for X_1 and X_2 is not needed since we know the optimal allocations and thus we need only the global budget w_0 .

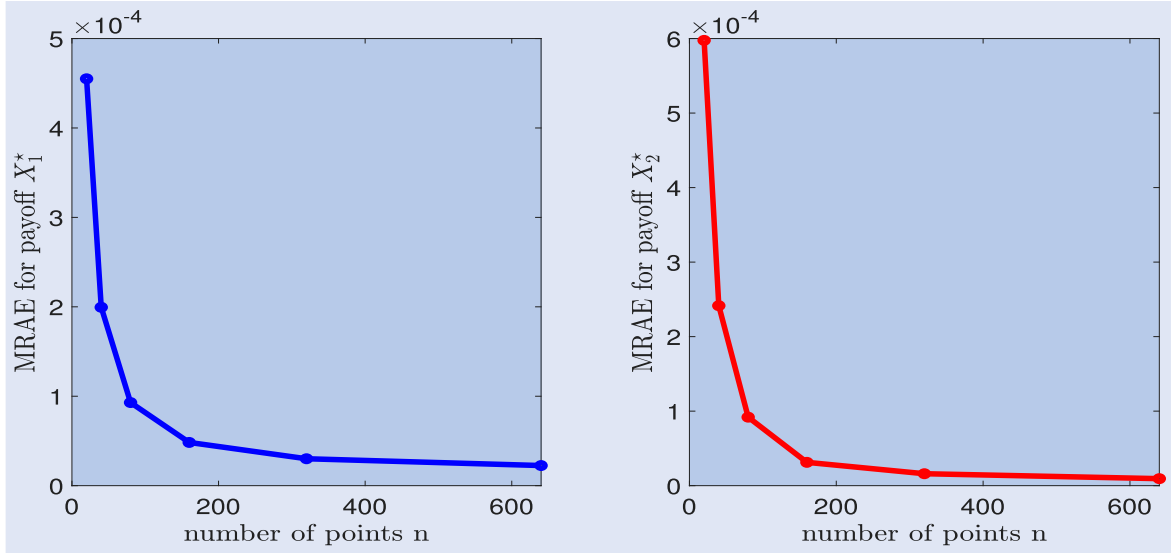


Figure 11. Mean relative absolute error in the case of weighted sum of two exponential utility functions using the GNum method. Input parameters: $\gamma_1 = 0.8$, $\gamma_2 = 0.2$, $k_1 = 0.3$, $k_2 = 0.7$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$ and $T = 1$.

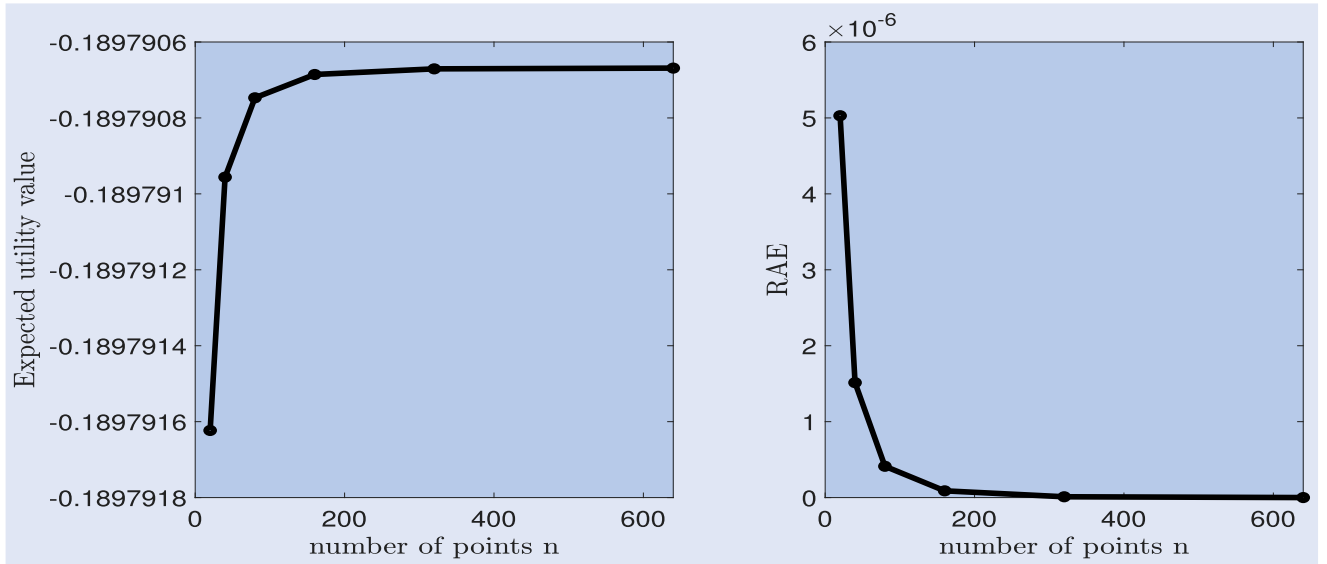


Figure 12. Expected utility value and relative absolute error in the case of a weighted sum of two exponential utility functions using the GNum method. Input parameters: $\gamma_1 = 0.8$, $\gamma_2 = 0.2$, $k_1 = 0.3$, $k_2 = 0.7$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$, and $T = 1$.

Now we implement the General Numerical approach in which we consider six payoffs $X_i, i = 1, \dots, 6$, and thus an expected utility defined as a weighted sum of six exponential utility functions.

Applying the algorithm for the GNum method with a number of runs $R = 5$, we get the outcome displayed in figure 15. The numerical optimal solutions obtained by means of the GNum method for payoffs $\tilde{X}_i^*, i = 1, \dots, 6$, with $n_5 = 640$ discretisation points are denoted by the pentagram marker, while the optimal payoffs $X_i^*, i = 1, \dots, 6$, computed using Proposition 4.4, are represented by solid lines; the GNum approach and the analytical solution provide a close match. Figure 16 exhibits the mean relative absolute error, observing a decreasing behaviour when the number of discretisation points increases, and figure 17 shows respectively the behaviour of the expected utility value and the relative absolute error as function of the discretisation points.

In the next section, we develop the case in which the objective function $V(\cdot)$ is expressed as a sum of distorted expectations, showing how to obtain the explicit solution and then applying the General Numerical approach to the case of a sum of two Range Value-at-Risk (RVaR).

5.2. Sum of distorted expectations with GNum

In Yaari's dual theory, an agent with terminal wealth X_T maximises the distorted expectation (also known as Yaari utility). Let $g : [0, 1] \rightarrow [0, 1]$, which is left-continuous and non-decreasing, satisfying $g(0) = 0$ and $g(1) = 1$, be the distortion function. Using the notation of Boudt *et al.* (2022) and assuming that g is absolutely continuous, the Yaari utility $\mathcal{H}_g(X_T)$ for an agent can be formulated as

$$\mathcal{H}_g(X_T) = \int_0^1 h(u) F_{X_T}^{-1}(u) du, \quad (24)$$

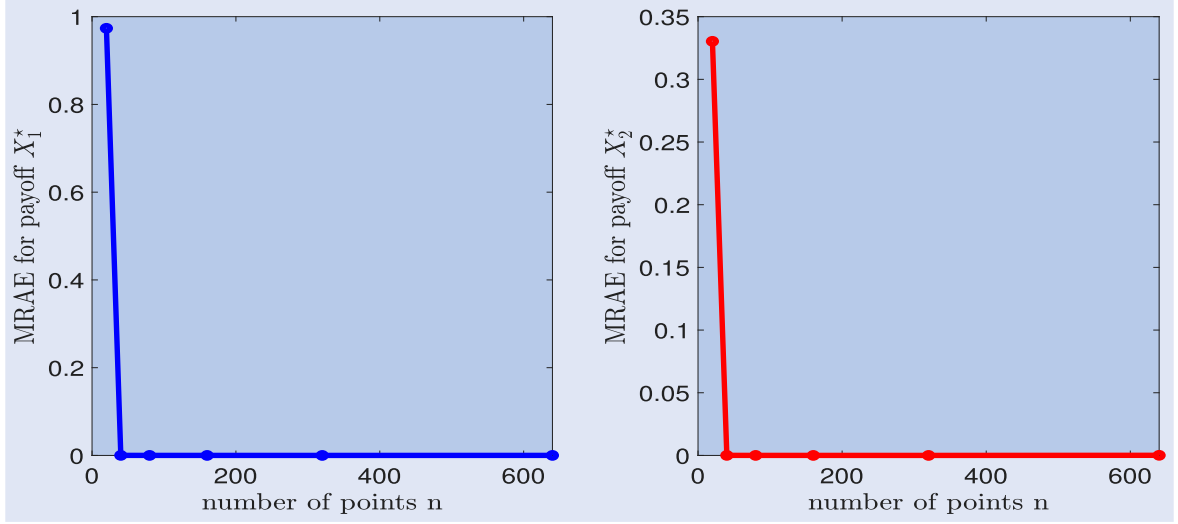


Figure 13. Mean relative absolute error in the case of weighted sum of two exponential utility functions using the GNum method (single cycle with $R = 5$ runs and starting 50-50 initial budget split). Input parameters: $\gamma_1 = 0.8$, $\gamma_2 = 0.2$, $k_1 = 0.3$, $k_2 = 0.7$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$ and $T = 1$.

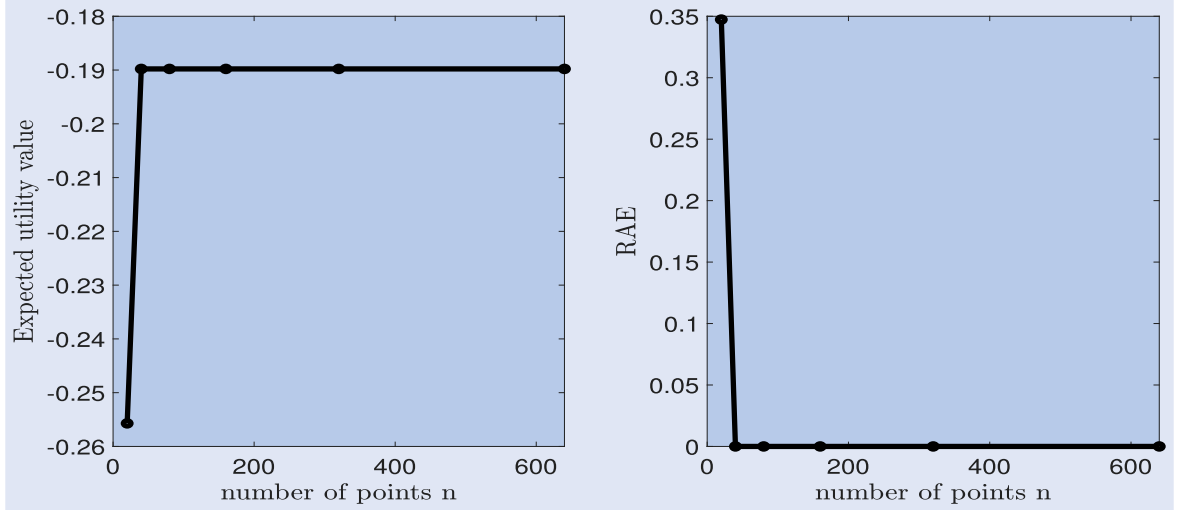


Figure 14. Expected utility value and RAE in the case of a weighted sum of two exponential utility functions using the GNum method (single cycle with $R = 5$ runs and starting 50-50 initial budget split). Input parameters: $\gamma_1 = 0.8$, $\gamma_2 = 0.2$, $k_1 = 0.3$, $k_2 = 0.7$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$ and $T = 1$.

where $h(u) := g'(1 - u)$, i.e. the left derivative of $g(x)$ in $1 - u$, is called the weighting function and satisfies $\int_0^1 h(u) du = 1$. As discussed in Boudt *et al.* (2022), the distortion function g plays a crucial role. Risk-averse behaviours are represented by a convex g function (e.g. Left Tail Value-at-Risk), while risk-seeking (also known as risk-loving) behaviours by a concave g function (e.g. Tail Value-at-Risk).

From this setting, the multivariate law-invariant objective function $V(\cdot)$ of problem (1) is defined as

$$V(X_1, \dots, X_d) := \sum_{i=1}^d \mathcal{H}_{g_i}(X_i), \quad (25)$$

where g_i refers to the individual distortion function for agent i , $i = 1, \dots, d$. For instance, in the case of preferences modelled by means of $\text{RVaR}_{\alpha, \beta}$ with $0 < \alpha < \beta \leq 1$, the individual

distortion and weighting functions are given by

$$g_i(u) = \min \left\{ \max \left\{ \frac{u + \beta_i - 1}{\beta_i - \alpha_i}, 0 \right\}, 1 \right\} \quad \text{and} \quad h_i(u) = \frac{1}{\beta_i - \alpha_i} \mathbb{1}_{(\alpha_i, \beta_i]}(u). \quad (26)$$

When the objective function $V(\cdot)$ is given by (25), the multivariate portfolio problem in (1) admits analytical solutions for payoffs $(X_1, \dots, X_d)$ at time T and the solution is direct and elementary because of Yaari utility's features.

In the case of a univariate portfolio choice problem under Yaari's dual theory framework, the optimal solution has been obtained in the literature by He and Jiang (2021) and Boudt *et al.* (2022). As pointed out in Boudt *et al.* (2022) the optimal payoff, provided that it exists and is anti-monotonic with the

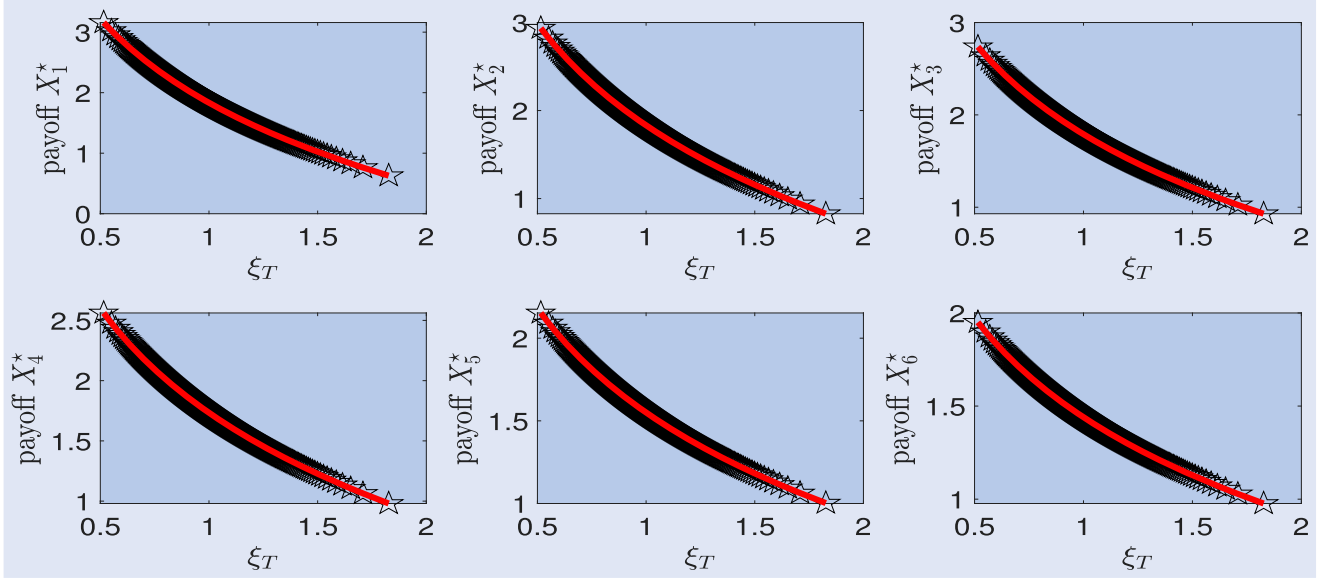


Figure 15. Optimal portfolios in the case of a weighted sum of six exponential utility functions using the GNum approach. Input parameters: $\gamma_1 = 0.5$, $\gamma_2 = 0.6$, $\gamma_3 = 0.7$, $\gamma_4 = 0.8$, $\gamma_5 = 1.1$, $\gamma_6 = 1.3$, $k_1 = k_2 = k_3 = k_4 = k_5 = k_6 = 1/6$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$, $T = 1$ and $n_5 = 640$.

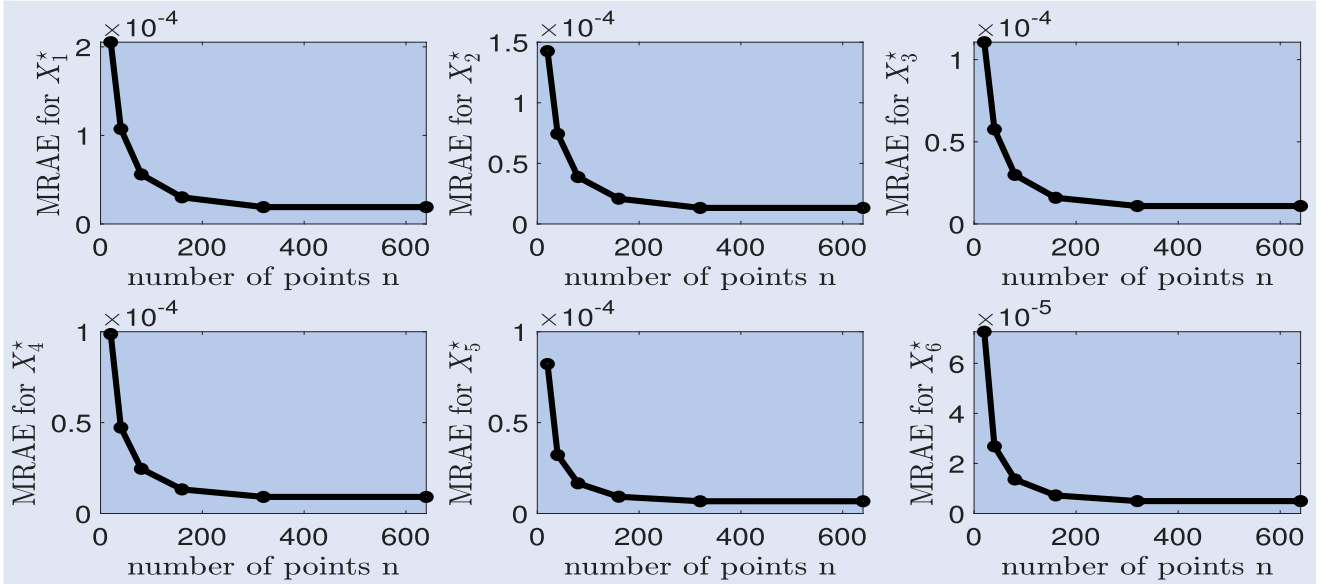


Figure 16. Mean relative absolute error in the case of a weighted sum of six exponential utility functions using the GNum approach. Input parameters: $\gamma_1 = 0.5$, $\gamma_2 = 0.6$, $\gamma_3 = 0.7$, $\gamma_4 = 0.8$, $\gamma_5 = 1.1$, $\gamma_6 = 1.3$, $k_1 = k_2 = k_3 = k_4 = k_5 = k_6 = 1/6$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$ and $T = 1$.

pricing kernel ξ_T , is characterised as a digital option or a risk-free investment. Such a result comes directly from the features of Yaari utility, which are comonotonic additivity and positive homogeneity (as observed by (24)).

Therefore, in a multivariate portfolio problem where $V(\cdot)$ is given by (25), exploiting the property of positive homogeneity, the problem boils down just to figure out which component in (25) yields to the highest value per unit of investment. Using such an approach, the problem reduces to finding the optimal payoff under a single Yaari utility for which solutions are available in the literature.

Section 5.2.1 illustrates the above framework with the special case of a sum of RVaRs when $d = 2$. We show the explicit

solution and then we use the GNum method to observe if the exact and numerical solutions match.

5.2.1. The case of the sum of two RVaRs. Let us recall the definition of Range Value-at-Risk (RVaR) proposed by Cont *et al.* (2010). For a confidence level $\alpha \in (0, 1)$, the Value-at-Risk (VaR) of X is the $100\alpha\%$ quantile of X ; formally, $\text{VaR}_\alpha(X) = F_X^{-1}(\alpha)$. From (26), it follows that RVaR can be expressed as

$$\text{RVaR}_{\alpha, \beta}(X) = \frac{1}{\beta - \alpha} \int_{\alpha}^{\beta} \text{VaR}_u(X) du, \quad 0 < \alpha < \beta \leq 1. \quad (27)$$

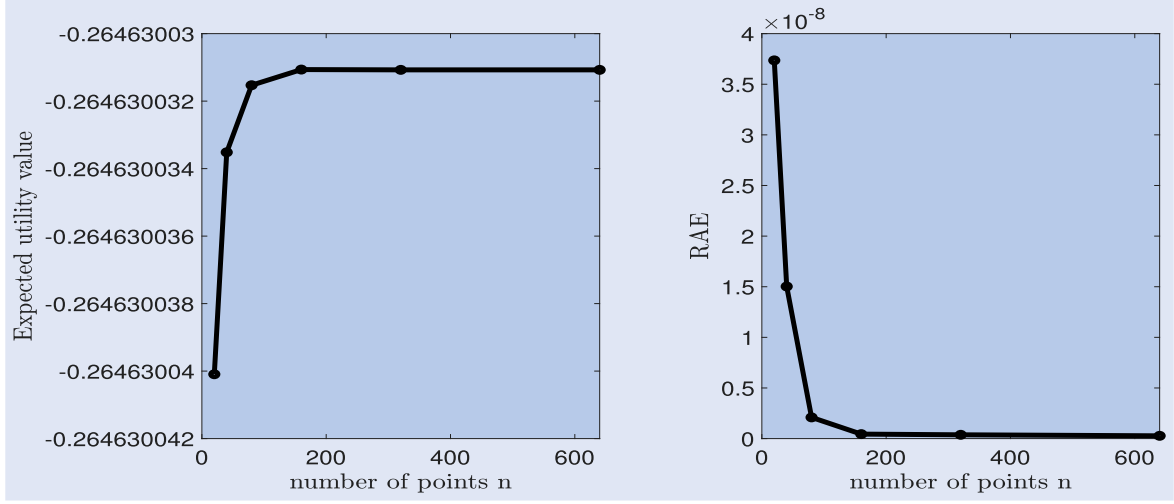


Figure 17. Expected utility value and relative absolute error in the case of a weighted sum of six exponential utility functions using the GNum approach. Input parameters: $\gamma_1 = 0.5$, $\gamma_2 = 0.6$, $\gamma_3 = 0.7$, $\gamma_4 = 0.8$, $\gamma_5 = 1.1$, $\gamma_6 = 1.3$, $k_1 = k_2 = k_3 = k_4 = k_5 = k_6 = 1/6$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$ and $T = 1$.

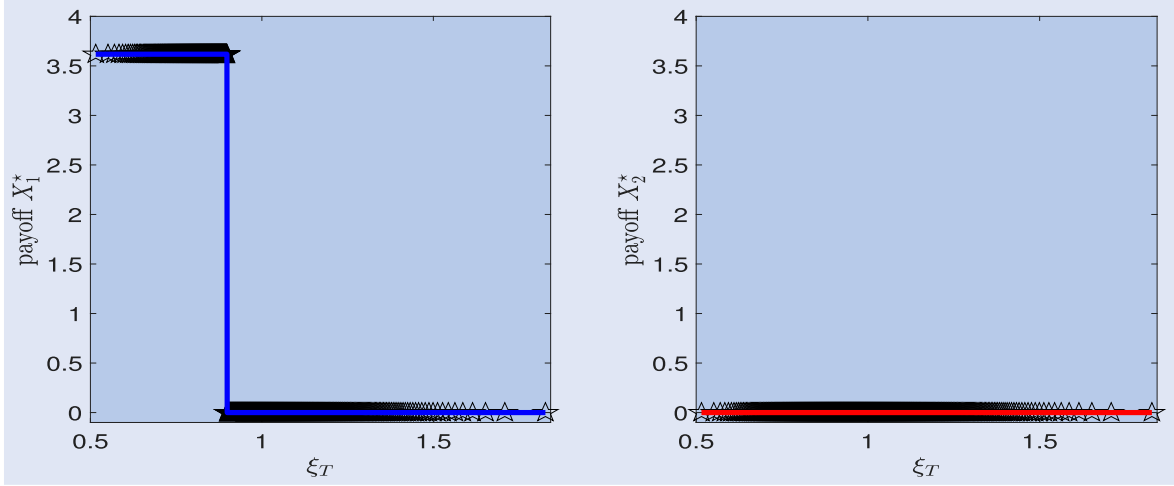


Figure 18. Optimal portfolios in the case of a sum of two RVaRs using the GNum approach. Input parameters: $\alpha_1 = 0.65$, $\beta_1 = 0.75$, $\alpha_2 = 0.6$, $\beta_2 = 0.9$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$, $T = 1$ and $n_5 = 640$.

In case of preferences modelled by RVaR, the objective function (25) when $d = 2$ is defined as

$$V(X_1, X_2) := \text{RVaR}_{\alpha_1, \beta_1}(X_1) + \text{RVaR}_{\alpha_2, \beta_2}(X_2). \quad (28)$$

Hereafter, we present an example in which the parameters of Agent 1 are $\alpha_1 = 0.65$ and $\beta_1 = 0.75$, while the parameters of Agent 2 are $\alpha_2 = 0.6$ and $\beta_2 = 0.9$. The pricing kernel ξ_T is defined as in (9) and $w_0 = 1$. Exploiting the findings of Boudt *et al.* (2022, Theorem 3.5), we get that the supremum of the Yaari ratio, $\sup_{c>0} \text{YR}(c)$, is greater than one for both agents. The maximum value of YR for Agent 1 is 3.581 with $c^* = 0.898$, while for Agent 2 the maximum value of YR is 3.075 with $c^* = 0.922$. The optimal payoff X_1^* and its Yaari utility have higher values with respect to the optimal payoff X_2^* and corresponding Yaari utility. Therefore, in a multivariate optimal portfolio choice problem, where the objective function is expressed as (28), there is no benefit in investing in payoff X_2 (due to the property of positive homogeneity), which leads to the outcome of figure 18. The optimal payoffs X_1^* and X_2^* for Agent 1 and Agent 2 are indicated respectively

by the solid blue and red lines; we observe a digital option behaviour for X_1^* and nothing for X_2^* .

Next we implement the GNum method developed in Section 5. To this end, we write (22) according to the appropriate choice of $f(\cdot)$ that depends on $V(\cdot)$ which is defined in (28). Accordingly, the optimisation problem can be stated as

$$\begin{aligned} & \sup_{(x_{i1}, z_i)_{i=1, \dots, n} \in \mathcal{A}'} \text{RVaR}_{\alpha_1, \beta_1}(x_{11}, \dots, x_{n1}) \\ & + \text{RVaR}_{\alpha_2, \beta_2}(z_1 - x_{11}, \dots, z_1 + \dots + z_n - x_{n1}) \end{aligned} \quad (29a)$$

for $i = 1, \dots, n$ and where the admissible set is now

$$\mathcal{A}' = \left\{ (x_{i1}, z_i) \in \mathbb{R} \times (\mathbb{R}_+), i = 1, 2, \dots, n \left| \sum_{i=1}^n \zeta_i z_i = w_0 \right. \right\} \quad (29b)$$

and $\zeta_i = \frac{1}{n} \sum_{k=i}^n \xi_k$, for $i = 1, \dots, n$.

Applying the algorithm for the GNum approach with $R = 5$, we get the numerical optimal payoffs $\tilde{X}_1^*$ and $\tilde{X}_2^*$,

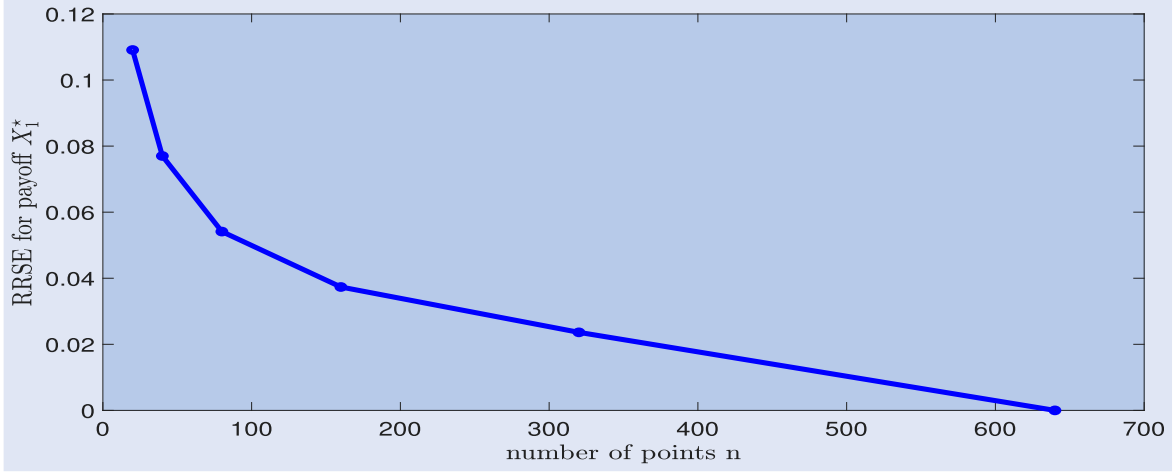


Figure 19. Root relative squared error for X_1^* in the case of a sum of two RVaRs using the GNum approach. Input parameters: $\alpha_1 = 0.65$, $\beta_1 = 0.75$, $\alpha_2 = 0.6$, $\beta_2 = 0.9$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$ and $T = 1$.

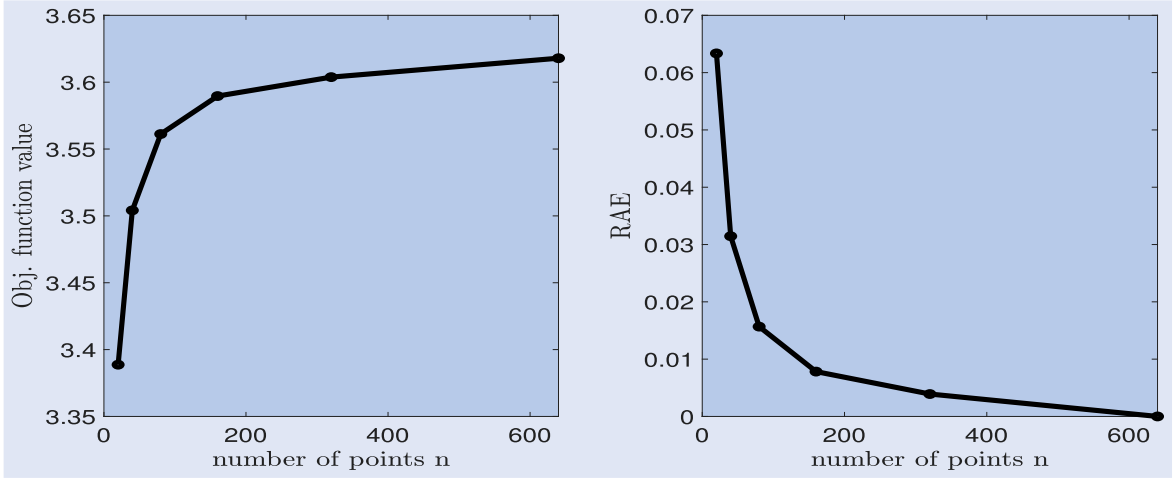


Figure 20. Value of the objective function in the case of a sum of two RVaRs using the GNum approach. Input parameters: $\alpha_1 = 0.65$, $\beta_1 = 0.75$, $\alpha_2 = 0.6$, $\beta_2 = 0.9$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$ and $T = 1$.

marked by the pentagram symbol, with $n_5 = 640$ discretisation points (see figure 18); the obtained numerical results overlap the optimal payoffs X_1^* and X_2^* . Figure 19 exhibits the root relative squared error[†] for X_1^* , showing a decreasing behaviour when the number of discretisation points increases and thus a convergence of the numerical optimal solution to the exact one. In the left panel of figure 20, the value of the objective function is displayed as a function of the number of discretisation points, whereas the right panel shows the relative absolute error between the objective function computed by means of the numerical approach and the one obtained through the analytical expression.

[†] Note that, to assess the accuracy of the GNum approach in this framework, the distance between the solution obtained numerically and the explicit solution is performed through the root relative squared error (RRSE), instead of MRAE (see Section 3.1), because some values are equal to zero. Let $\tilde{x}_{jk}^*$ and x_{jk}^* , for $j = 1, \dots, n$, be respectively the optimal solutions of the numerical approach and the optimal solutions of the analytical one. Then, the RRSE, for $r = 0, 1, \dots, R$ and $k = 1, \dots, d$, is defined as $EP_{r,k}^R := \sqrt{\sum_{j=1}^n (\tilde{x}_{jk}^* - x_{jk}^*)^2 / \sum_{j=1}^n (x_{jk}^*)^2}$.

6. Optimal portfolios without closed-form solutions

In this section, we present two examples for which, to the best of our knowledge, no other method appears to be available in the existing literature. These cases illustrate, in principle, that the GNum approach can be used to address optimisation problems that remain unsolved by current techniques. Let us consider the first example:

$$V(X_1, X_2) = -\frac{1-\gamma_1}{\gamma_1} \left(\frac{X_1}{1-\gamma_1} + b_1 \right)^{\gamma_1} \frac{1-\gamma_2}{\gamma_2} \times \left(\frac{X_2}{1-\gamma_2} + b_2 \right)^{\gamma_2}.$$

As the risk-sharing problem (3) has no known solution, the MtU approach to solve the portfolio choice problem (1) cannot be applied. In fact, to the best of our knowledge, no method is available in the literature to solve it. However, we can still apply the GNum approach. Figure 21 displays the numerical optimal payoffs $\tilde{X}_1^*$ and $\tilde{X}_2^*$.

We point out that they are not comonotonic (Bernard et al. 2023 show, however, that optimality requires that $\tilde{X}_1^* + \tilde{X}_2^*$ is anti-monotonic with respect to ξ_T).

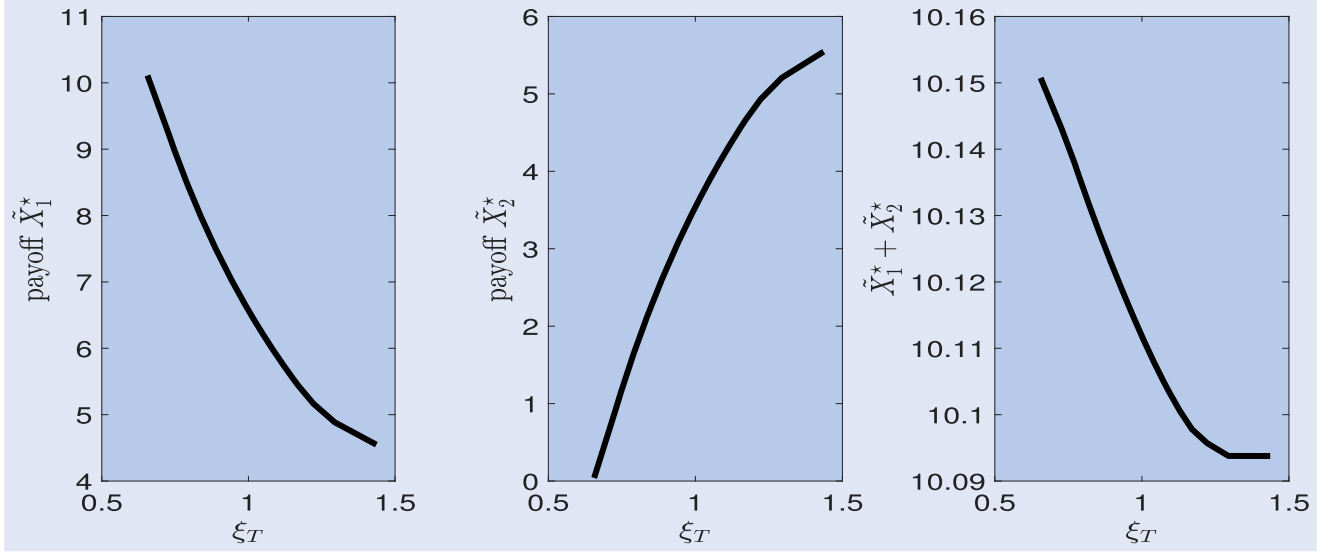


Figure 21. Left: $\tilde{X}_1^*$ wrt ξ_T . Center: $\tilde{X}_2^*$ wrt ξ_T . Right: $\tilde{X}_1^* + \tilde{X}_2^*$ wrt ξ_T . Both $\tilde{X}_1^*$ and $\tilde{X}_2^*$ are obtained using the GNum approach. Input parameters: $\gamma_1 = 0.4$, $b_1 = 0.2$, $\gamma_2 = 0.6$, $b_2 = -0.1$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$, $T = 1$ and $w_0 = 10$.

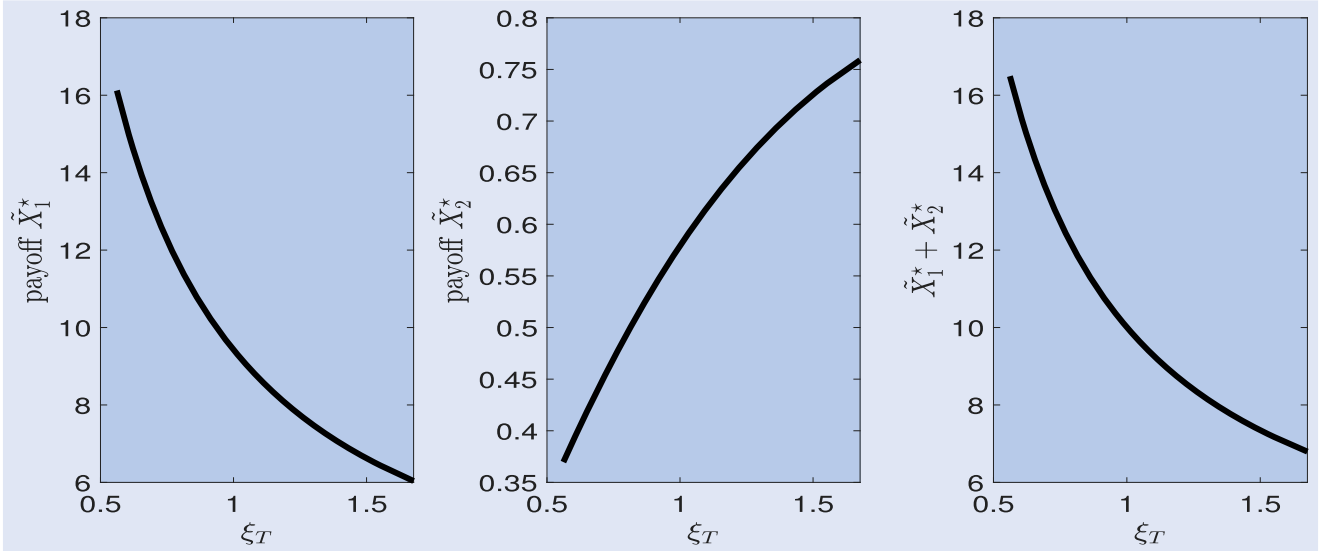


Figure 22. Left: $\tilde{X}_1^*$ wrt ξ_T . Center: $\tilde{X}_2^*$ wrt ξ_T . Right: $\tilde{X}_1^* + \tilde{X}_2^*$ wrt ξ_T . Both $\tilde{X}_1^*$ and $\tilde{X}_2^*$ are obtained using the GNum approach. Input parameters: $\gamma_1 = 1$, $\gamma_2 = 0.5$, $b = 0.4$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$, $T = 1$ and $w_0 = 10$.

We now present the second example characterised by the following objective function:

$$V(X_1, X_2) = \gamma_1 \log(X_1) + \gamma_2 \log(X_2) - b\sqrt{X_1 X_2}.$$

By applying the GNum approach we obtain the numerical optimal payoffs $\tilde{X}_1^*$ and $\tilde{X}_2^*$, which are shown in figure 22.

These examples thus show that solutions to multivariate optimal portfolio choice problems do not have to exhibit comonotonicity per se.

7. Conclusions

This paper proposes new methods to solve multivariate optimal portfolio selection problems. Under some conditions, we show how to reduce the dimensionality of the problem

at hand to become a univariate quantile optimisation problem that can be solved by an accurate and fast numerical method (MtU approach). Second, when these conditions are not satisfied, we are still able to propose a general algorithm (GNum approach) that solves the multivariate portfolio selection problem numerically. When applying this approach on some problems for which closed-form expressions exist (such as the maximisation of a weighted sum of expected utilities), we observe that it performs reasonably well. To further demonstrate its applicability, we also consider two examples for which, to best of our knowledge, no other approach seems to be available in the literature. These examples suggest that the GNum approach could be applied in such cases.

Disclosure statement

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Appendices

Appendix 1. Proofs

A.1. Proof of Proposition 4.1

Let $w_{i,0} \in \mathbb{R}$, for $i = 1, \dots, d$, be the budget allocation of payoff X_i at time zero such that $\sum_{i=1}^d w_{0,i} = w_0$. Then we can formulate problem (1), for $i = 1, \dots, d$, as follows:

$$\max_{(X_1, \dots, X_d)} \mathbb{E}[\mathcal{U}(X_1, \dots, X_d)], \quad \text{s.t. } \mathbb{E}[\xi_T X_i] = w_{0,i}. \quad (\text{A1})$$

Introducing d Lagrange multipliers, i.e. $\lambda_1, \dots, \lambda_d > 0$ and using pathwise optimisation, then we rewrite the problem in terms of an auxiliary problem. Specifically, for each $\omega \in \Omega$, we have

$$\max_{(x_1, \dots, x_d)} \left\{ \mathcal{U}(x_1, \dots, x_d) - \sum_{i=1}^d \lambda_i \xi_T(\omega) x_i \right\}. \quad (\text{A2})$$

Given the assumptions stated in Proposition 4.1 regarding the utility function $\mathcal{U}$, the optimal solutions $(x_1^*, \dots, x_d^*)$, which are found by taking the first-order conditions (hereafter also FOCs), are at $\nabla(\mathcal{U}(x_1, \dots, x_d) - \sum_{i=1}^d \lambda_i \xi_T(\omega) x_i) = \mathbf{0}$ where $\mathbf{0}$ is a vector of zeros. Specifically, $(x_1^*, \dots, x_d^*) = ([k_1 U_1']^{-1}(\lambda_1 \xi_T(\omega)), \dots, [k_d U_d']^{-1}(\lambda_d \xi_T(\omega)))$. As done in Bernard *et al.* (2018), for each $\omega \in \Omega$ and for $i = 1, \dots, d$, we define the random variable $X_{\lambda_i}^*(\omega)$ as follows.

$$X_{\lambda_i}^*(\omega) := [k_i U_i']^{-1}(\lambda_i \xi_T(\omega)),$$

and we choose a $\lambda_i^* > 0$ such that $\mathbb{E}[\xi_T X_{\lambda_i}^*] = w_{0,i}$. The conditions imposed on U_i and by continuity of the mapping $\lambda_i \mapsto \mathbb{E}[\xi_T X_{\lambda_i}^*]$ assure the unique existence of a λ_i^* . Therefore, for every vector of the final payoff $(X_1, \dots, X_d)$ that satisfies the budget constraint, we have by construction that

$$\begin{aligned} & \mathcal{U}(X_1(\omega), \dots, X_d(\omega)) - \sum_{i=1}^d \lambda_i^* \xi_T(\omega) X_i(\omega) \\ & \leq \mathcal{U}(X_{\lambda_1}^*(\omega), \dots, X_{\lambda_d}^*(\omega)) - \sum_{i=1}^d \lambda_i^* \xi_T(\omega) X_{\lambda_i}^*(\omega) \quad \forall \omega \in \Omega, \end{aligned} \quad (\text{A3})$$

given that the vector $(X_{\lambda_1}^*(\omega), \dots, X_{\lambda_d}^*(\omega))$ is the optimal solution to (A2). Finally, we take the expectation on both sides of the inequality (A3) obtaining thus

$$\mathbb{E}[\mathcal{U}(X_1, \dots, X_d)] \leq \mathbb{E}[\mathcal{U}(X_{\lambda_1}^*, \dots, X_{\lambda_d}^*)]. \quad (\text{A4})$$

Any budget allocation $w_{0,i}$, for $i = 1, \dots, d$, which satisfies the total budget constraint $\sum_{i=1}^d w_{0,i} = w_0$ can give different solutions of (A1). To obtain the solution of problem (1), we maximise

$$\max_{(w_{0,1}, \dots, w_{0,d}) \in [0, w_0]^d} \mathbb{E}[\mathcal{U}(X_{\lambda_1}^*, \dots, X_{\lambda_d}^*)], \quad \text{s.t. } \sum_{i=1}^d w_{0,i} = w_0. \quad (\text{A5})$$

The solution of (A5) exists, i.e. the vector $(w_{0,1}^*, \dots, w_{0,d}^*)$, since the optimisation is computed for a continuous function over a set of $\mathbb{R}^d$.

A.2. Proof of Proposition 4.3

The proof follows immediately from Proposition 4.1. Let $w_{0,1}$ and $w_{0,2}$ be the allocated budget at time zero of respectively X_1 and X_2 such that $w_{0,1} + w_{0,2} = w_0$ (i.e. the global budget). Then let λ_1, λ_2 be the Lagrange multipliers and, for each $\omega \in \Omega$, consider the following auxiliary problem

$$\max_{(x_1, x_2)} \{k_1 U_{\gamma_1}(x_1) + k_2 U_{\gamma_2}(x_2) - \lambda_1 \xi_T(\omega) x_1 - \lambda_2 \xi_T(\omega) x_2\}. \quad (\text{A6})$$

Taking the first-order conditions, the optimal solution is the form of

$$\begin{pmatrix} X_1^* \\ X_2^* \end{pmatrix} = \begin{pmatrix} -\ln\left(\frac{\lambda_1 \xi_T(\omega)}{k_1 \gamma_1}\right) / \gamma_1 \\ -\ln\left(\frac{\lambda_2 \xi_T(\omega)}{k_2 \gamma_2}\right) / \gamma_2 \end{pmatrix}, \quad (\text{A7})$$

where $\lambda_1, \lambda_2 > 0$ are such that the budget equations are $\mathbb{E}[\xi_T X_1^*] = w_{0,1}$ and $\mathbb{E}[\xi_T X_2^*] = w_{0,2}$. Then, inserting (A7) into the budget equations, we obtain

$$\begin{aligned} \lambda_1^* &= k_1 \gamma_1 \exp\left(-\gamma_1 e^{rT} w_{0,1} + rT - \frac{\theta^2 T}{2}\right), \\ \lambda_2^* &= k_2 \gamma_2 \exp\left(-\gamma_2 e^{rT} w_{0,2} + rT - \frac{\theta^2 T}{2}\right). \end{aligned} \quad (\text{A8})$$

The last step consists in finding the optimal split of the budget $(w_{0,1}^*, w_{0,2}^*)$ such that $w_{0,1}^* + w_{0,2}^* = w_0$. Thus, for some $\lambda \in \mathbb{R}$, $w_{0,1} = w_0 \lambda$ and $w_{0,2} = w_0(1 - \lambda)$. Maximising $\mathbb{E}[k_1 U_{\gamma_1}(X_1^*)] + \mathbb{E}[k_2 U_{\gamma_2}(X_2^*)]$ over λ and making calculations, we obtain the optimal allocation coefficient λ^* such that there is a unique budget split $w_{0,1}^* = w_0 \lambda^*$ and $w_{0,2}^* = w_0(1 - \lambda^*)$. Plugging expression (A8) with the right budget split in (A7) and rearranging yields (16).

A.3. Proof of Proposition 4.4

The proof follows the same arguments of proof of Proposition 4.3, extended to d dimensions. The only difference consists in finding the optimal split of the budget $(w_{0,1}^*, w_{0,2}^*, \dots, w_{0,d}^*)$ such that $\sum_{i=1}^d w_{0,i}^* = w_0$. Let $\lambda_i \in \mathbb{R}$ be the allocation coefficient assigned to X_i , then $w_{0,i} = w_0 \lambda_i$ for $i = 1, \dots, d$. Then we maximise $\sum_{i=1}^d \mathbb{E}[k_i U_{\gamma_i}(X_i^*)]$ over $\lambda_1, \dots, \lambda_d$. After substituting and making calculations, we get the optimal allocation coefficients $\lambda_1^*, \dots, \lambda_d^*$ such that the budget split is given by $w_{0,i}^* = w_0 \lambda_i^*$ for $i = 1, \dots, d$.

A.4. Proof of Proposition 4.5

We show the proof of optimal allocations in a general form. Starting from (12) we have $k_i \gamma_i \exp(-\gamma_i \tilde{X}_i) = c$, for $i = 1, \dots, d$, where c is a constant. Rearranging, we obtain

$$\tilde{X}_i = \frac{\ln(k_i \gamma_i)}{\gamma_i} - \frac{\ln(c)}{\gamma_i}. \quad (\text{A9})$$

Taking the sum over i and recalling that $\sum_{i=1}^d X_i = S$, we get

$$S = \sum_{i=1}^d \frac{\ln(k_i \gamma_i)}{\gamma_i} - \sum_{i=1}^d \frac{1}{\gamma_i} \ln(c). \quad (\text{A10})$$

Let $\frac{1}{\gamma} = \frac{1}{\gamma_1} + \dots + \frac{1}{\gamma_d}$, then

$$-\ln(c) = \gamma S - \gamma \sum_{i=1}^d \frac{\ln(k_i \gamma_i)}{\gamma_i}. \quad (\text{A11})$$

We substitute (A11) in (A9) to obtain the optimal allocations:

$$\tilde{X}_j = \frac{\gamma}{\gamma_j} S + \frac{\ln(k_j \gamma_j)}{\gamma_j} - \frac{\gamma}{\gamma_j} \sum_{i=1}^d \frac{\ln(k_i \gamma_i)}{\gamma_i}, \quad j = 1, \dots, d, \quad (\text{A12})$$

which ends the proof.

For further details regarding a similar proof in case of an exponential utility function $U_{\gamma_i}(x) = 1/\gamma_i[1 - \exp(-\gamma_i x)]$, we refer to Gerber and Pafumi (1998, Example 7, p. 81).

Appendix 2. Case of weighted sum of CRRA utility functions

In this appendix, we repeat the study of the MtU approach performed with exponential utility functions (Section 4.1) in the context of a weighted sum of two CRRA utility functions. Specifically, we consider the same collective utility function proposed in (13) where now the univariate utility function $U_{\gamma_i}(X_i)$ is defined as

$$U_{\gamma_i}(x) = \begin{cases} \ln(x) & \gamma_i = 1 \\ \frac{x^{1-\gamma_i}}{1-\gamma_i} & \gamma_i \neq 1, \gamma_i > 0 \end{cases}. \quad (\text{A13})$$

The parameter γ_i is the risk-aversion coefficient of the CRRA utility function.

We present an analytical solution in the case of a weighted sum of two CRRA utility functions (Proposition A.1) when the pricing kernel is defined as in Section 4 and then the optimal allocations. The numerical analysis of the MtU method in the case of homogeneous and heterogeneous preferences is presented respectively in Appendices A.5 and A.6.

Proposition A.1 provides the explicit solution, by means of Proposition 4.1, of a weighted sum of two CRRA utility functions when preferences are heterogeneous.

PROPOSITION A.1 *The a.s. unique optimal solution (X_1^*, X_2^*) to the problem*

$$\max_{(X_1, X_2) \in \mathcal{A}} \mathbb{E}[k_1 U_{\gamma_1}(X_1) + k_2 U_{\gamma_2}(X_2)], \quad (\text{A14})$$

where $\mathcal{A} := \{(X_1, X_2) : \mathbb{E}[\xi_T(X_1 + X_2)] = w_0\}$ is given by

$$\begin{pmatrix} X_1^* \\ X_2^* \end{pmatrix} = \begin{pmatrix} \left(\frac{\xi_T \lambda_1^*}{k_1} \right)^{-\frac{1}{\gamma_1}} \\ \left(\frac{\xi_T \lambda_2^*}{k_2} \right)^{-\frac{1}{\gamma_2}} \end{pmatrix}, \quad (\text{A15})$$

where

$$\begin{aligned} \lambda_1^* &= k_1 \left(\frac{\exp\left(\left(rT + \frac{\theta^2 T}{2\gamma_1}\right)\left(\frac{1}{\gamma_1} - 1\right)\right)}{\lambda^* w_0} \right)^{\gamma_1}, \\ \lambda_2^* &= k_2 \left(\frac{\exp\left(\left(rT + \frac{\theta^2 T}{2\gamma_2}\right)\left(\frac{1}{\gamma_2} - 1\right)\right)}{(1 - \lambda^*) w_0} \right)^{\gamma_2}, \end{aligned} \quad (\text{A16})$$

and λ^* is the unique solution of the following equation:

$$\frac{k_1 \exp\left(\left(rT + \frac{\theta^2 T}{2\gamma_1}\right)(1 - \gamma_1)\right) w_0^{1-\gamma_1}}{k_2 \exp\left(\left(rT + \frac{\theta^2 T}{2\gamma_2}\right)(1 - \gamma_2)\right) w_0^{1-\gamma_2}} = \frac{\lambda^{\gamma_1}}{(1 - \lambda)^{\gamma_2}}. \quad (\text{A17})$$

Proof The proof follows immediately from Proposition 4.1 and details can be found in Bernard et al. (2023, see Appendix B). Using the same arguments of the above proof and taking the FOCs, the optimal solution is

$$\begin{pmatrix} X_1^* \\ X_2^* \end{pmatrix} = \begin{pmatrix} \left(\frac{\lambda_1 \xi_T(\omega)}{k_1} \right)^{-\frac{1}{\gamma_1}} \\ \left(\frac{\lambda_2 \xi_T(\omega)}{k_2} \right)^{-\frac{1}{\gamma_2}} \end{pmatrix}, \quad (\text{A18})$$

where $\lambda_1, \lambda_2 > 0$ are such that the budget equations are $\mathbb{E}[\xi_T X_1^*] = w_{0,1}$ and $\mathbb{E}[\xi_T X_2^*] = w_{0,2}$. Then, we insert the optimal solution

(X_1^*, X_2^*) into the budget equations obtaining so

$$\begin{aligned} \lambda_1^* &= \frac{k_1 \exp\left\{\left(rT + \frac{\theta^2 T}{2\gamma_1}\right)(1 - \gamma_1)\right\}}{w_{0,1}^{\gamma_1}}, \\ \lambda_2^* &= \frac{k_2 \exp\left\{\left(rT + \frac{\theta^2 T}{2\gamma_2}\right)(1 - \gamma_2)\right\}}{w_{0,2}^{\gamma_2}}. \end{aligned} \quad (\text{A19})$$

Then we need to find the optimal split of the budget $(w_{0,1}^*, w_{0,2}^*)$ such that $w_{0,1}^* + w_{0,2}^* = w_0$. Recall that $w_{0,1} = w_0 \lambda$ and $w_{0,2} = w_0(1 - \lambda)$ for some $\lambda \in (0, 1)$. As done previously, we maximise $\mathbb{E}[k_1 U_{\gamma_1}(X_1^*)] + \mathbb{E}[k_2 U_{\gamma_2}(X_2^*)]$ with respect to λ obtaining thus unique λ^* that satisfies (A17). λ^* can be solved numerically and there is a unique budget split $w_{0,1}^* = w_0 \lambda^*$ and $w_{0,2}^* = w_0(1 - \lambda^*)$. Inserting (A19) with the right budget split in (A18) and rearranging leads to the solution of Proposition A.1, which ends the proof. ■

The implementation of the Multivariate-to-Univariate approach within the CRRA framework necessitates the availability of optimal allocations. These are provided in Gerber and Pafumi (1998) and summarised in Proposition A.2 in the case of homogeneous preferences. Remark A.1 considers the case of heterogeneous preferences.

PROPOSITION A.2 *The optimal risk-sharing $(X_1, X_2, \dots, X_d)$ of S , $i = 1, \dots, d$, for homogeneous CRRA utility functions is given by*

$$X_i := Y_i(S) = q_i S, \quad (\text{A20})$$

where $q_i = k_i^{1/\gamma_i} / \sum_{j=1}^d k_j^{1/\gamma_j}$.

Proof See Example 9 of Gerber and Pafumi (1998, Section 8, p. 82). ■

REMARK A.1 It is not always possible to derive closed-form expressions for optimal allocations such as is the case of CRRA utility functions in the presence of heterogeneous risk aversion coefficients. However, optimal allocations can be retrieved exploiting numerical procedures. Let us consider the utility function expressed in (A13) when $\gamma_i \neq 1$. Specifically, $U_{\gamma_i}(x) = \frac{x^{1-\gamma_i}}{1-\gamma_i}$ for $i = 1, \dots, d$. From (12), we obtain $k_i \check{X}_i^{-\gamma_i} = c$ or $\check{X}_i = k_i^{1/\gamma_i} c^{-1/\gamma_i}$ where c is just a constant. As made in Proof A.4 and recalling that $\sum_{i=1}^d X_i = S$, we sum over i obtaining so $S = \sum_{i=1}^d k_i^{1/\gamma_i} c^{-1/\gamma_i}$. Such a solution exists numerically and reads as follows:

$$\text{solve } c \in \mathbb{R}_+ \text{ such that } S = \sum_{i=1}^d k_i^{1/\gamma_i} c^{-1/\gamma_i}. \quad (\text{A21})$$

Similar to the numerical analysis presented in Section 4.1, and with reference to Proposition A.2 and Remark A.1, we implement the MtU approach in the context of homogeneous preferences (Appendix A.5) and heterogeneous preferences (Appendix A.6). The input parameters are displayed directly in the figure captions. We recall that the weights k_1 and k_2 are chosen arbitrarily and the global budget is $w_0 = 10$.

A.5. Weighted sum of two homogeneous CRRA utility functions

Given that optimal allocations of CRRA utility functions under homogeneous preferences are known in closed-form expressions (see Proposition A.2), we proceed to implement the Multivariate-to-Univariate approach.

Applying the algorithm for the MtU method with a number of runs $R = 5$, we obtain the numerical optimal payoffs $\check{X}_1^*$ and $\check{X}_2^*$, which are denoted by the marker symbol $*$. As shown in figure A1, the outcomes obtained with 640 discretisation points match closely the

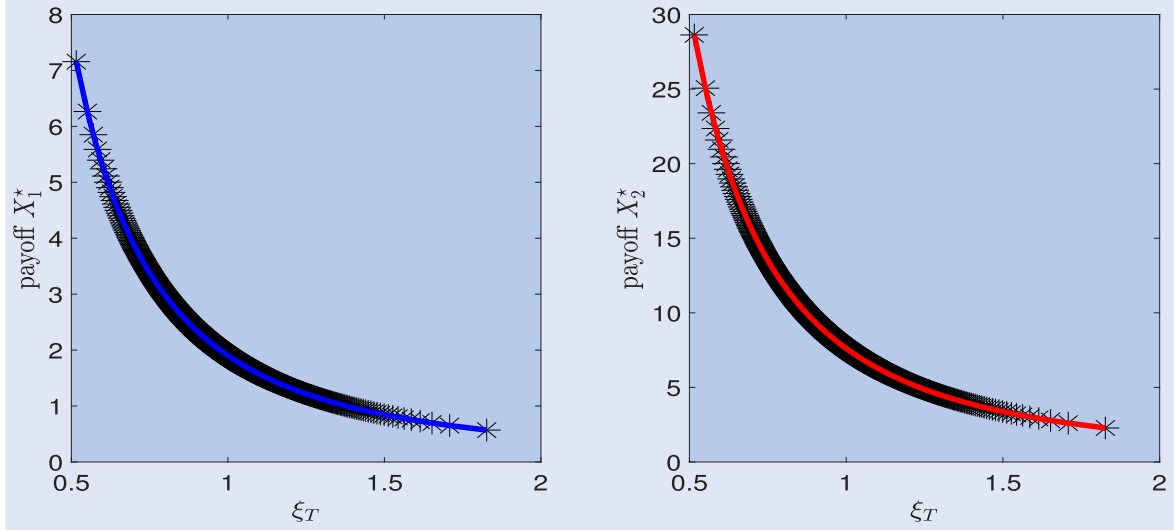


Figure A1. Optimal portfolios in the case of a weighted sum of two CRRA utility functions using the MtU approach. Input parameters: $\gamma_1 = \gamma_2 = 0.5$, $k_1 = 1/3$, $k_2 = 2/3$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$, $T = 1$ and $n_5 = 640$.

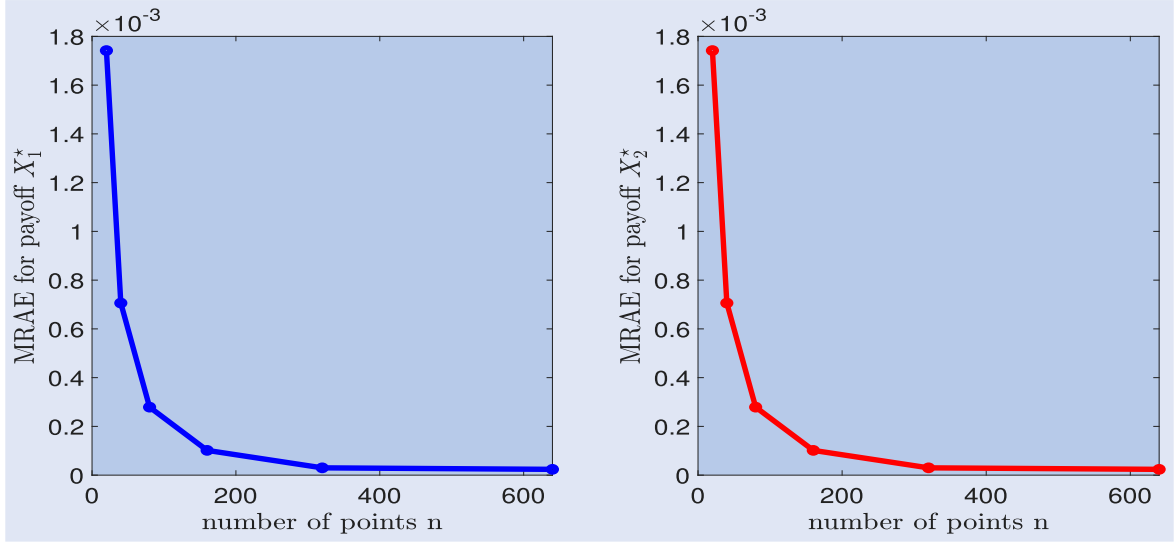


Figure A2. MRAE in the case of a weighted sum of two CRRA utility functions using the MtU approach. Input parameters: $\gamma_1 = \gamma_2 = 0.5$, $k_1 = 1/3$, $k_2 = 2/3$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$ and $T = 1$.

optimal payoffs X_1^* (solid blue line) and X_2^* (solid red line) computed through Proposition A.1.

Figure A2 displays the mean relative absolute error between the numerical and analytical solutions for payoffs X_1^* and X_2^* . Both pictures point out that the MRAE decreases as the number of points increases, highlighting thus the convergence of optimal solutions to the exact ones.

In figure A3, we report respectively the expected utility value computed by means of numerical optimal solutions and its relative absolute error (see Section 3.2) when the number of points grows up.

A.6. Weighted sum of two heterogeneous CRRA utility functions

In this section, we present the Multivariate-to-Univariate approach for a weighted sum of two CRRA utility functions when the coefficients of risk aversion are different; e.g. $\gamma_1 = 0.3$ and $\gamma_2 = 0.7$.

The possibility of using the MtU method is feasible since optimal allocations can be obtained through the numerical solution stated in Remark A.1. From the algorithm with $R = 5$, the numerical optimal payoffs $\tilde{X}_1^*$ and $\tilde{X}_2^*$ are displayed in figure A4 (see marker symbol $*$) and match the optimal payoffs X_1^* (solid blue line) and X_2^* (solid red line). In figure A5, we plot the mean relative absolute error as a function of discretisation points and we observe that the MRAE for both payoffs decreases when the number of points increases.

Figure A6 displays the value of the collective utility function $\mathcal{U}$ (left panel) and its relative absolute error (right panel). Both figures A5 and A6 show once again the validity of the Multivariate-to-Univariate approach.

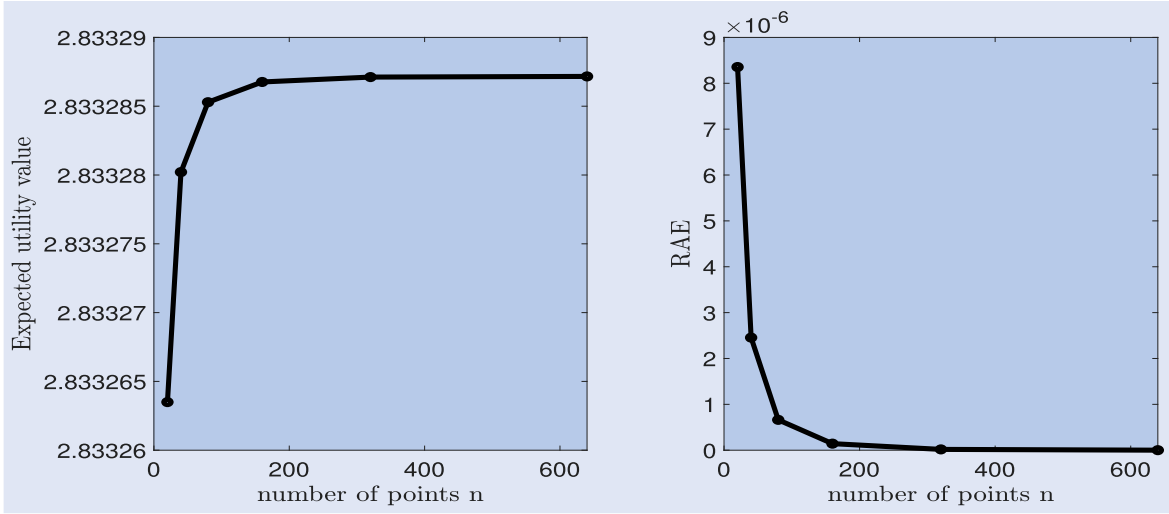


Figure A3. Utility value and relative absolute error in the case of a weighted sum of two CRRA utility functions using the MtU approach. Input parameters: $\gamma_1 = \gamma_2 = 0.5$, $k_1 = 1/3$, $k_2 = 2/3$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$ and $T = 1$.

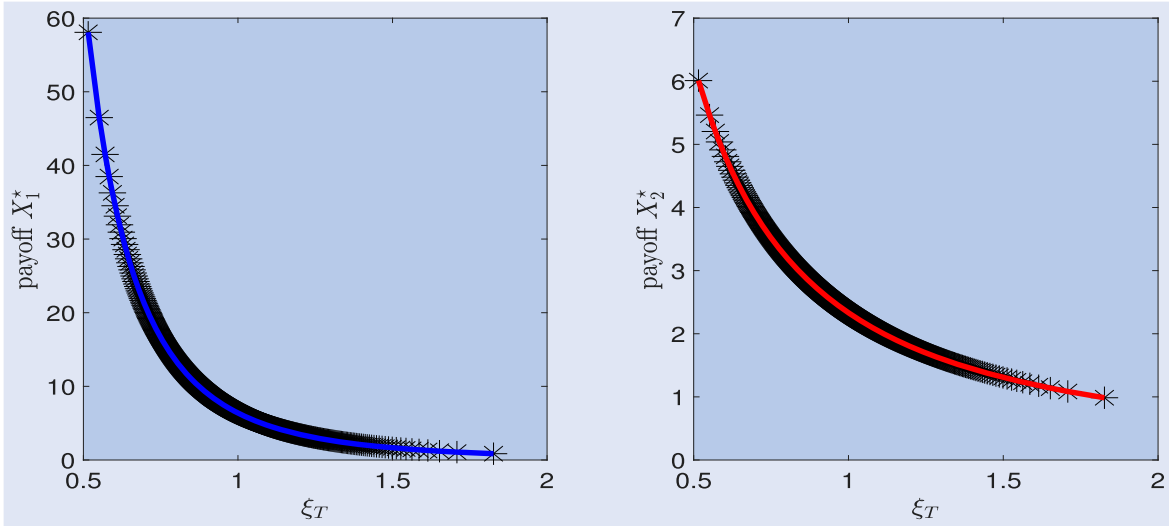


Figure A4. Optimal portfolios in the case of a weighted sum of two heterogeneous CRRA utility functions using the MtU approach. Input parameters: $\gamma_1 = 0.3$, $\gamma_2 = 0.7$, $k_1 = 0.49$, $k_2 = 0.51$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$, $T = 1$ and $n_5 = 640$.

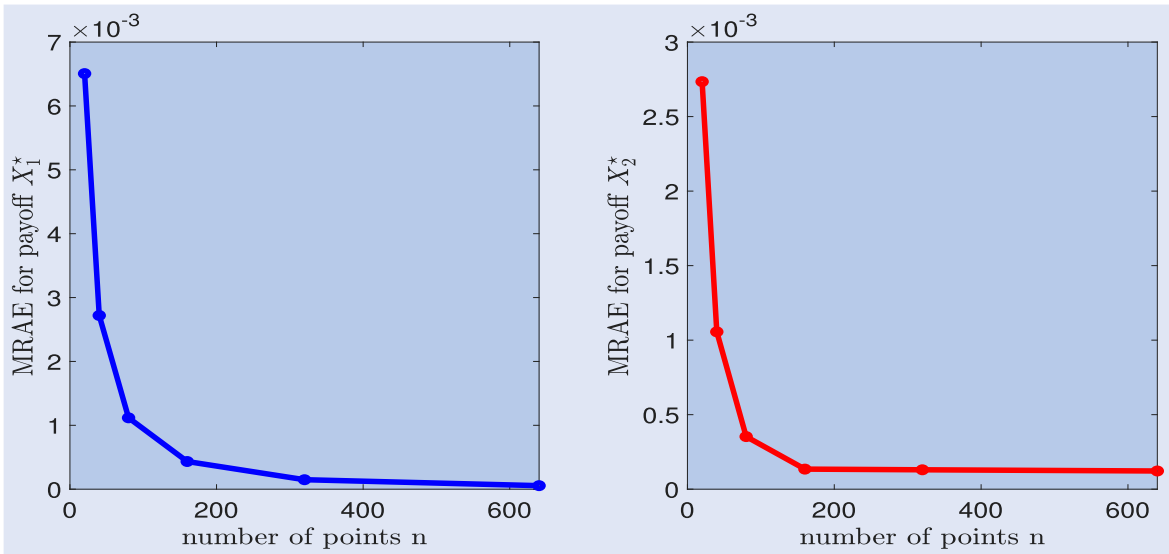


Figure A5. MRAE in the case of a weighted sum of two heterogeneous CRRA utility functions using the MtU approach. Input parameters: $\gamma_1 = 0.3$, $\gamma_2 = 0.7$, $k_1 = 0.49$, $k_2 = 0.51$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$ and $T = 1$.

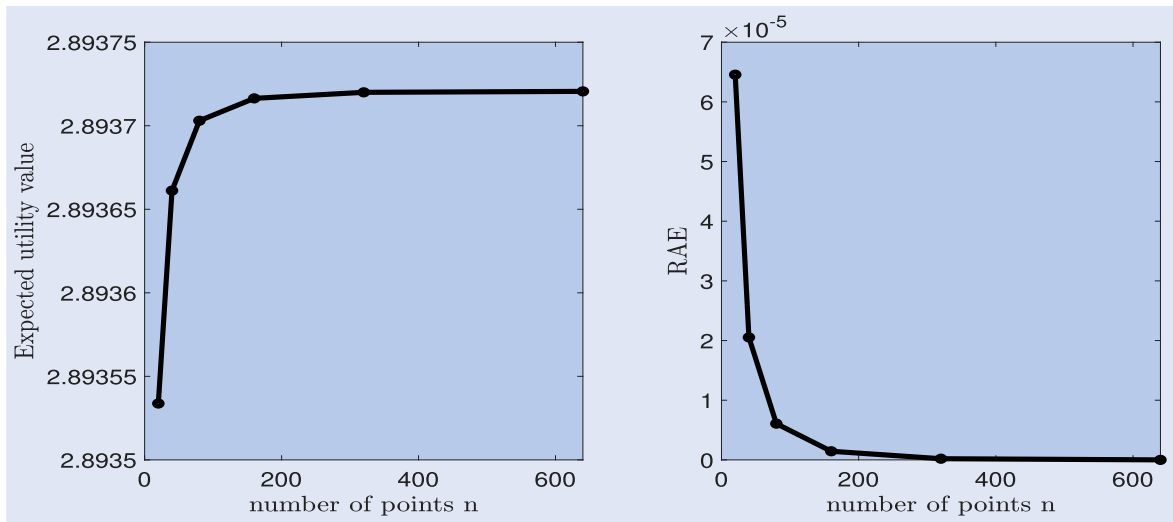


Figure A6. Utility value and relative absolute error in the case of a weighted sum of two heterogeneous CRRA utility functions using the MtU approach. Input parameters: $\gamma_1 = 0.3$, $\gamma_2 = 0.7$, $k_1 = 0.49$, $k_2 = 0.51$, $\mu = 0.05$, $r = 0.01$, $\sigma = 0.2$ and $T = 1$.